

Chronic Kidney Disease Progression in Autosomal Dominant Polycystic Kidney Disease

Statistics Comprehensive Group Project

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Abstract

Autosomal dominant polycystic kidney disease (ADPKD) is the most prevalent monogenic cause of end-stage kidney disease (ESKD), yet population-level longitudinal data from South Asian cohorts remain sparse. Established prognostic instruments were developed predominantly in Western or East Asian populations with preserved early renal function, limiting their applicability to Indian patients who commonly present at advanced disease stages.

We conducted a single-centre retrospective longitudinal cohort study of 193 patients with ADPKD followed across up to 26 outpatient visits. The primary estimand, the annualized rate of eGFR decline, was obtained by patient-level ordinary least squares (OLS) regression of eGFR on visit number, with an assumption that two

successive visits are at regular intervals. Secondary outcomes included a composite renal endpoint of $\geq 40\%$ eGFR decline or ESKD, Kaplan-Meier event-time analysis with log-rank testing, and a multivariable OLS regression with nine standardized predictors.

At baseline, the cohort was markedly advanced: median eGFR was $19.0 \text{ mL/min/1.73 m}^2$ (IQR 10.0-45.9), 41.3% had CKD stage G5, and 87.7% were hypertensive. The median annualized eGFR slope was $-1.73 \text{ mL/min/1.73 m}^2/\text{year}$. In multivariable OLS, lower baseline eGFR, older age, and a history of urinary tract infections (UTIs) were independent predictors, explaining 9.0% of slope variance ($R^2 = 0.090$). A statistically significant Age $\times$ baseline eGFR interaction was also identified.

This Indian ADPKD cohort presents at a disease stage substantially more advanced than any published international series, reflecting late nephrology referral. The counter-intuitive positive UTI-slope association is best explained by survivor-selection bias and ascertainment artefact. These findings motivate structured Indian ADPKD registries, family screening programmes, and future prospective studies incorporating total kidney volume imaging and genetic profiling.

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1 Introduction

Autosomal dominant polycystic kidney disease (ADPKD) is the most common hereditary kidney disease, estimated to affect between 1 in 400 and 1 in 1,000 individuals worldwide and responsible for 5–10% of ESKD cases globally (Torres et al., 2007). The disease is caused by loss-of-function mutations in *PKD1* (chromosome 16p; $\approx 85\%$ of cases, encoding polycystin-1) or *PKD2* (chromosome 4q; $\approx 15\%$ of cases, encoding polycystin-2) (Harris and Torres, 2009). Bilateral renal cysts expand inexorably, displacing normal parenchyma, activating the renin–angiotensin–aldosterone system, and ultimately producing CKD that progresses to ESKD in approximately 50% of patients by the sixth decade of life.

1.1 Clinical Heterogeneity and Established Risk Factors

The ADPKD phenotype is profoundly heterogeneous. Even within families carrying identical causative mutations, individuals exhibit wide variability in total kidney volume (TKV) growth, eGFR decline rate, and age at ESKD onset. Established determinants of rapid progression include: (i) male sex; (ii) PKD1 truncating mutations; (iii) hypertension before age 35 years; (iv) early-onset macroscopic haematuria; (v) recurrent urinary tract infections (UTIs); and (vi) increasing height-adjusted TKV (Cornec-Le Gall et al., 2016; Oh et al., 2023). The Mayo Imaging Classification (MIC), stratifying patients into classes 1A–1E based on htTKV growth rate, has emerged as the pre-eminent prognostic instrument and serves as the primary eligibility criterion for tolvaptan therapy in most regulatory jurisdictions (?).

1.2 The South Asian Evidence Gap

Despite this substantial literature, nearly all prognostic data derive from Western cohorts or large East Asian registries. In India, ADPKD is frequently diagnosed late—often only after significant renal impairment has already accrued—and epidemiological data from Indian cohorts remain sparse (?). This evidence gap has profound implications: risk scores calibrated in earlier-stage populations may poorly discriminate in cohorts where the majority of patients already have CKD G4–G5 at first presentation. Furthermore, the comorbidity burden, dietary patterns, and healthcare access in India differ substantially from reference populations, potentially modulating ADPKD progression through distinct pathways.

1.3 Some Important Biomarkers

Longitudinally monitoring a selective panel of clinical and biochemical biomarkers is essential to characterize the progression pathways and systemic manifestations of Autosomal

Dominant Polycystic Kidney Disease (ADPKD) insight into physiological processes ranging from filtration decline to metabolic imbalances:

- **Serum Creatinine (S_{cr}):** A cornerstone metabolic waste product that serves as the primary indicators of baseline renal clearance capacity and parenchymal filtration damage.
- **Serum Electrolytes (Sodium and Potassium):** Indispensable markers used to track fluid regulation and homeostatic stability, which are critical for capturing clinical complications such as hyperkalemia as functional nephron mass diminishes.
- **Haemoglobin (Hb):** Tracked as a vital indicator of clinical anemia, which frequently arises from a progressive decline in renal endocrine function and subsequent erythropoietin deficiency in advanced cystic disease.
- **Proteinuria and Urine ACR:** Indicators of glomerular structural disruption and macromolecular leakage that function as strong predictors of accelerated disease trajectories.

Estimated Glomerular Filtration Rate (eGFR) is the major biomarker used in the study and is computed using the 2021 CKD-EPI creatinine equation:

$$eGFR = 142 \times \min(S_{cr}/\kappa, 1)^\alpha \times \max(S_{cr}/\kappa, 1)^{-1.200} \times 0.9938^{\text{Age}} \times [1.012 \text{ if Female}]$$

where

- S_{cr} : Serum creatinine (mg/dL),
- κ : 0.7 for females and 0.9 for males,
- α : -0.241 for females and -0.302 for males, and
- **Age**: Patient age in years.

1.4 Study Objectives and Pre-Specified Research Questions

The present study provides a detailed longitudinal statistical analysis of ADPKD progression from a single-centre Indian cohort. We address the following pre-specified research questions:

- Q1** What is the **risk and timing** of reaching the composite endpoint of End-Stage Kidney Disease (ESKD), or $> 40\%$ eGFR decline, or dialysis, or transplantation from the baseline visit? What are the effects of other factors on it?
- Q2** What is the **annualised rate of eGFR decline** in this Indian ADPKD cohort? How does it depend on the various external factors?

- Q3** What is the relative contribution of hypertension, proteinuria and baseline kidney function to the rate of CKD progression in ADPKD?
- Q4** What is the prevalence and **severity distribution** (Mild/Moderate/Severe) of anemia in this cohort, and how does it correlate with biological sex and progressive CKD staging?

2 Data and Study Design

2.1 Data Preparation

This research comprises a single-center, retrospective longitudinal cohort study conducted under the clinical oversight of **Dr. Pratim Sengupta**, a distinguished Nephrologist, and Renal Transplant Physician, who is also the founder of Nephro Care India Limited. Data management and project coordination were facilitated by **Mr. Chinmay Hazra** throughout the study’s duration.

The study population was established through the consecutive enrollment of patients with a confirmed diagnosis of Autosomal Dominant Polycystic Kidney Disease (ADPKD), with clinical data captured through systematic serial outpatient follow-ups.

To facilitate longitudinal modelling, a significant structural transformation of the dataset was required. The primary data was originally captured in a “wide format,” wherein individual patient visits ($V_1, V_2, \dots, V_{26}$) were recorded as distinct columnar variables. This was subsequently restructured into a “long-format” (person-visit) analytic framework, ensuring the dataset was optimally configured for further analysis.

2.2 Inclusion and Exclusion

The study cohort was restricted to patients aged 18 years or older at the baseline visit, with a diagnosis of Autosomal Dominant Polycystic Kidney Disease (ADPKD) confirmed through established imaging criteria. Following initial enrollment, participants were monitored across a longitudinal trajectory comprising between 1 and 26 clinical encounters.

The final analytic dataset was refined to ensure clinical relevance. Specifically, visit-level exclusion was strictly based on the simultaneous absence of both **Serum Creatinine (Cr)** and **Estimated Glomerular Filtration Rate (eGFR)** measurements. Conversely, any observation was retained if at least one of these primary renal biomarkers was recorded. Given that **eGFR** serves as the primary metric for the study’s inferential objectives, this exclusion protocol ensures that the longitudinal analysis is based strictly on informative clinical data points. The final analytic cohort comprised 193 unique patients contributing 982 longitudinal observations across up to 26 visits.

2.3 Variable Definition and and Missingness Profile

To provide a comprehensive overview of the longitudinal ADPKD dataset, we present a unified master schema (Table 1). This table integrates clinical definitions, biological reference ranges, and statistical types with their corresponding data presence percentages. Variables are organized hierarchically into four analytical tiers based on their availability, ranging from the “Near Complete” (Tier 1) to “Sparse” (Tier 4).

Table 1: Unified Master Schema: Definitions, Reference Ranges, and Data Presence

Variable Name	Type	Presence	Clinical Definition and Context
Tier 1: Near-Complete(> 95%)			
SL_No	Numerical	100%	Unique Patient Identifier; used for panel indexing.
Gender	Categorical	100%	Biological sex (M/F). <i>Context: Baseline covariate.</i>
Age	Continuous	100%	Patient age (years). <i>Context: Primary predictor of eGFR decline.</i>
Visit_Num	Ordinal	100%	Integer visit sequence (V_1 - V_{26}) for longitudinal tracking.
Cr	Continuous	99.9%	Serum Creatinine (mg/dL) - Muscle metabolism waste. Healthy: 0.7-1.3 (M), 0.6-1.1 (F). <i>Context: Primary Renal Biomarker.</i>
HTN	Categorical	99.7%	Hypertension status (yes/no). <i>Context: Key comorbidity and exposure variable.</i>
eGFR	Continuous	97.6%	Estimated GFR (mL/min/1.73m ²) via 2021 CKD-EPI. Healthy: > 90; Failure: < 15. <i>Context: Primary Outcome Variable.</i>
Tier 2: High-Availability (75% – 95%)			

Variable Name	Type	Presence	Clinical Definition and Research Context
K	Continuous	86.2%	Serum Potassium (mmol/L). Healthy range: 3.5-5.0. <i>Context: Electrolyte balance indicator.</i>
Hb	Continuous	80.7%	Hemoglobin (g/dL) - RBC oxygen-carrying protein. Healthy: 13.5-17.5 (M), 12.0-15.5 (F). <i>Context: Clinical Anemia Marker.</i>
Na	Continuous	77.9%	Serum Sodium (mmol/L). Healthy range: 135-145. <i>Context: Fluid/Electrolyte balance marker.</i>

Tier 3: Low-to-Moderate-Availability (30% – 75%)

Urine_pus	Categorical	61.3%	Presence of WBCs in urine. <i>Context: Indicator of Urinary Tract Infection.</i>
Urine_ACR	Continuous	48.8%	Albumin-to-Creatinine Ratio. <i>Context: Quantitative measure of glomerular damage.</i>
Ca	Continuous	44.6%	Serum Calcium (mg/dL). Healthy range: 8.5-10.2. <i>Context: Mineral/Bone metabolism predictor.</i>
PO4	Continuous	41.2%	Serum Phosphate (mg/dL). Healthy range: 2.5-4.5. <i>Context: Mineral metabolism indicator.</i>
SGPT / SGOT	Continuous	~ 38%	Alanine/Aspartate Aminotransferase. <i>Context: Hepatic function markers.</i>
Uric_Acid	Continuous	37.4%	Serum Uric Acid (mg/dL). <i>Context: Metabolic comorbidity marker.</i>

Variable Name	Type	Presence	Clinical Definition and Research Context
Urine_Pr_Cleaned	Ordinal	34.3%	Normalized Proteinuria Score (0.0 to 4.0). <i>Context: Standardized feature for mediation analysis.</i>
Tier 4: Sparse (< 30%)			
UTI / CVD	Categorical	~ 26%	History of infection or Cardiovascular Disease.
iPTH	Continuous	26.5%	Parathyroid Hormone. <i>Context: Endocrine control of mineral levels.</i>
Transplant	Categorical	24.9%	History of Renal Transplant or Dialysis.
Vasc_Calc	Numerical	21.3%	Vascular Calcification Score based on imaging.
Cyst_haem	Categorical	14.3%	Indicator of Cyst Haemorrhage. <i>Note: Systematic missingness in V_1 by design.</i>

3 Data Cleaning and Preprocessing

Longitudinal clinical data collected in routine nephrology practice are inherently susceptible to entry errors, definitional inconsistencies, and structured missingness that arises from the natural heterogeneity of follow-up intensity across patients. Prior to any inferential analysis, a principled and reproducible data-cleaning protocol was applied to the raw dataset. The decisions taken at each stage are documented below, together with their clinical and statistical justification.

3.1 Recoding of Categorical Variables

Hypertension (HTN): The hypertension variable was recorded in free text across the data-collection period. The majority of entries were binary (**yes/no**), but a small number of records included a duration qualifier (e.g., **yes (4 years)**). Because the duration

annotation was present in only a negligible fraction of observations, and because the analytical objective requires a binary indicator rather than a time-since-diagnosis variable, all duration-qualified entries were recoded to 1 (hypertension present). This conservative decision ensures that no true hypertensive patient is misclassified as normotensive. Patient 45 had no hypertension record in any visit; this patient was assigned a missing indicator and excluded from hypertension-stratified sub-analyses.

3.2 Identification and Handling of Typographic Errors

Two unambiguous data-entry errors were identified through physiological plausibility screening.

Patient 14, Visit 2 - Serum Potassium: The recorded potassium value for this visit was 4 Jan, 2026, which is evidently a date string entered into a numeric field, consistent with a copy-paste error originating from an adjacent date column. The value was replaced with a system missing indicator and subsequently imputed via the procedure described in Section 3.8.

Patient 60, Visit 1 - Serum Sodium: A sodium value of 9.4 mEq/L was recorded. The physiological range compatible with life is approximately 125–165 mEq/L; a value of 9.4 mEq/L is incompatible with survival and is conclusively typographic in origin (most plausibly a missing leading digit, i.e., intended as 139.4 mEq/L or similar). The value was set to missing and imputed.

3.3 Correction of Column-Shift Error

For Patient 13 at Visit 1, a systematic rightward column-shift of one field position was detected in the raw data. Specifically, the values for Urine Albumin-to-Creatinine Ratio (ACR), Serum Potassium (K), and Serum Sodium (Na) were each displaced one column to the right of their correct positions. The error was identified by cross-referencing the field labels against the magnitude and units of the recorded values (e.g., a value of 4.2 appearing in the Na field, a value consistent with K rather than Na). The three values were re-assigned to their correct columns. No other visit for this patient was recorded, confirming this was an isolated single-visit data-entry anomaly.

3.4 Exclusion of the Urine Pus Biomarker

Despite having a relatively high data presence of approximately 61.3% (602 out of 982 longitudinal observations), the `Urine_pus` column was excluded from downstream inferential and predictive modeling due to severe, systemic recording errors and unstructured heterogeneity. A detailed inspection of the raw fields revealed a highly problematic mixture

of disparate reporting formats, ranging from unstructured qualitative descriptions (e.g., “plenty”, “occasional”, “few”, and “nil”) to semi-quantitative numeric intervals (e.g., “10 to 15”, “0-2”, and “30-35”), along with formula corruption flags (e.g., #NAME?). Most critically, the column suffered from pervasive date-conversion artifacts and column-shift anomalies, which mistakenly introduced fully-formed date strings and timestamps (e.g., “01-02-2026 00:00” and “01-Feb”) directly into the numeric field. Because these multiple structural corruptions cannot be reliably mapped onto a coherent continuous scale or standard ordinal thresholds without introducing extreme bias, the variable was deemed statistically uninterpretable and omitted from the final analytic dataset.

3.5 Standardization of Semi-Quantitative Proteinuria

To maximize the analytical utility of the available clinical records, the Urine Protein (`Urine_Pr`) variable underwent a multi-stage standardization procedure to standardize the highly heterogeneous reporting formats. In its primary state, the raw data exhibited significant unstructured variance, comprising a mixture of semi-quantitative descriptors (e.g., “Trace”, “++”), quantitative numeric values with inconsistent units, and non-specific qualitative annotations. The cleaning process transformed these disparate entries into a unified, five-point ordinal scale (0 to 4), which effectively expanded the subset of observations available for correlation analysis from a nominal 6 to an informative 340 data points.

Linguistic standardization was first applied to normalize free-text entries for case sensitivity and terminological redundancy. Following the conversion of all records to lowercase to ensure consistency, a clinical mapping was applied: terms such as “nil”, “negative”, “-ve”, and “absent” were mapped to an ordinal score of 0, while descriptions of “trace” or “faint trace” were assigned a value of 0.5. Semi-quantitative “plus” markers were directly translated into their numeric equivalents, where markers ranging from 1+ to 4+ were mapped to the ordinal values 1.0 through 4.0, respectively.

For observations recorded as quantitative numeric values, regular expression (Regex) parsing was utilized to extract digits and identify the associated units of measurement. Unit normalization was performed by standardizing all mass-based entries to milligrams; specifically, records in grams (e.g., 2.46 g) were converted via a factor of 1,000. To ensure these numeric values were comparable with the clinician-reported semi-quantitative scale, they were binned according to standard nephrology thresholds. Values below 10 mg were classified as 0 (Negative); 10-30 mg as 0.5 (Trace); 30-100 mg as 1 (1+); 100-300 mg as 2 (2+); 300-1,000 mg as 3 (3+); and values exceeding 1,000 mg were mapped to 4 (4+).

Finally, the analytic dataset was shielded from pathological outliers and non-informative noise that would otherwise distort downstream association coefficients. A single entry

of 46,083 was identified as a non-physiological anomaly—likely a result of a clerical unit error or a catastrophic entry failure—and was subsequently excluded from the analysis. Furthermore, ambiguous descriptive entries that lacked a clear quantifiable magnitude (e.g., clinical notes indicating the presence of bacteria without protein quantification) were set to a system-missing indicator (NaN). This disciplined filtering ensures that the resulting ordinal variable represents a coherent clinical metric suitable for robust correlation and longitudinal modelling.

3.6 Imputation: Necessity and Analytical Rationale

Imputation in this study is not undertaken merely to obtain a complete rectangular dataset. It serves a specific and non-trivial analytical function: the primary inferential models - mixed-effects regression for longitudinal eGFR trajectory and Cox proportional-hazards regression for time to renal replacement therapy-require complete covariate vectors for all observations used in estimation. Listwise deletion under these frameworks would not only reduce effective sample size but, more critically, would introduce bias whenever the missing data mechanism is not missing completely at random (MCAR).

Serum creatinine, potassium, sodium, and haemoglobin are direct inputs to the trajectory models, not merely auxiliary descriptors. Their absence from particular visits is clinically patterned: values are more likely to be missing at early, lower-acuity visits when not all biochemical panels are ordered, and less likely to be missing at advanced-stage visits when full metabolic panels become routine. This pattern is consistent with a missing-at-random (MAR) mechanism conditional on disease severity—precisely the setting under which multiple imputation yields valid inference. Imputation is therefore analytically required: it preserves the longitudinal structure of each patient’s record, allows the full mixed-effects likelihood to be evaluated, and prevents the differential exclusion of visits that would otherwise distort estimated progression trajectories.

3.7 Multiple Imputation by Chained Equations (MICE)

Missing values in Serum Creatinine (Cr), Serum Sodium (Na), Serum Potassium (K), and Haemoglobin (Hb) were addressed using Multiple Imputation by Chained Equations (MICE). In the MICE framework, each incomplete variable is imputed by a separate conditional model, treating all other variables as predictors, and the process cycles iteratively until convergence. The predictors selected for each target variable were Age and Gender, both of which have well-established physiological relationships with the respective biomarkers: age influences renal function, electrolyte homeostasis, and erythropoietic activity; gender determines haemoglobin reference ranges and modulates creatinine production through muscle mass differences.

Algorithm 1 provides a formal statement of the procedure as implemented in Python.

Algorithm 1: Multiple Imputation by Chained Equations (MICE)

Input : Dataset $\mathbf{X}$ with missing values

Output: Imputed datasets $\{\mathbf{X}^{(m)}\}_{m=1}^M$

```

1 Initialize missing values in  $\mathbf{X}$  with initial estimates to get  $\mathbf{X}^{(0)}$ ;
2 for  $m = 1$  to  $M$  do
3    $\mathbf{X}^{(m)} \leftarrow \mathbf{X}^{(0)}$ ;
4   for each variable  $j$  with missing values do
5     Let  $\mathbf{X}_{\neg j}$  be the dataset excluding variable  $j$ ;
6     Let  $\mathbf{X}_j^{\text{obs}}$  be the observed values of variable  $j$ ;
7     Let  $\mathbf{X}_{\neg j}^{\text{obs}}$  be the corresponding observed values in  $\mathbf{X}_{\neg j}$ ;
8     Fit a regression model  $f_j$  on  $(\mathbf{X}_{\neg j}^{\text{obs}}, \mathbf{X}_j^{\text{obs}})$ ;
9     Predict the missing values  $\mathbf{X}_j^{\text{miss}}$  using  $f_j$  and  $\mathbf{X}_{\neg j}^{\text{obs}}$ ;
10    Replace the missing values in  $\mathbf{X}_j$  with these predictions;
11  Store  $\mathbf{X}^{(m)}$ ;
12 return  $\{\mathbf{X}^{(m)}\}_{m=1}^M$ ;

```

MICE exploits the joint conditional distribution of the incomplete variables without requiring a parametric multivariate normality assumption on the full data. By conditioning on Age and Gender, each imputed draw respects the biological dependency structure: a high-creatinine imputed value for an elderly male is more probable than for a young female, mirroring the actual physiology. The generation of $M > 1$ completed datasets propagates imputation uncertainty into all downstream point estimates and standard errors, yielding confidence intervals and p -values that are valid under the MAR assumption.

However, MICE operates by minimising a statistical loss function and does not impose hard physiological boundaries on imputed values. Because the algorithm balances across all variables simultaneously, it can, in principle, generate imputed values that are statistically consistent with the predictor space yet clinically implausible—for example, a serum sodium value of 155 mEq/L paired with a simultaneously imputed potassium of 7.2 mEq/L, a combination that is physiologically untenable. This limitation motivates the use of Predictive Mean Matching as the imputation method within the MICE framework, as described below.

3.8 Predictive Mean Matching (PMM)

To address the boundary problem inherent to unconstrained MICE, all imputations were performed using Predictive Mean Matching (PMM) as the elemental imputation method within each chained equation. PMM constrains imputed values to lie within the empirical range of the observed data, thereby guaranteeing that no imputed value is physiologically

impossible by construction.

Algorithm 6 describes the PMM procedure applied to each incomplete variable at each iteration of the MICE cycle.

Algorithm 2: Predictive Mean Matching (PMM): The Clinical Donor Strategy

Input : Target patient with missing biomarker y_{miss} and predictors $\mathbf{x}_{\text{miss}}$;

Reference set of patients with observed biomarker y_{obs} and predictors $\mathbf{X}_{\text{obs}}$

Output: Imputed value y^*

- 1 Fit a regression model f using the observed data pool: $y_{\text{obs}} \sim f(\mathbf{X}_{\text{obs}})$;
 - 2 Calculate predicted values (proxy scores) $\hat{y}_i = f(\mathbf{x}_i)$ for all individuals;
 - 3 Identify a donor pool $\mathcal{D}$ consisting of k patients whose predicted scores $\hat{y}_{\text{obs}}$ are closest to the target’s predicted score $\hat{y}_{\text{miss}}$;
 - 4 Randomly select a single donor d from the candidate pool $\mathcal{D}$;
 - 5 Assign the donor’s actual measured value to the missing entry: $y^* \leftarrow y_d$;
 - 6 **return** y^* ;
-

The decisive property of PMM is that the imputed value y^* is always an *observed* value drawn from the real data. This ensures that the imputed dataset cannot contain values outside the physiological range present in the observed cohort, resolving the boundary problem of unconstrained MICE.

In the present application, serum creatinine values imputed by PMM are guaranteed to lie within the range observed in the cohort (0.49 to 19.1 mg/dL); potassium values within 2.5–7.2 mEq/L; sodium within the cohort-observed range; and haemoglobin within the sex-stratified range observed in the data. This preserves both the clinical plausibility and the longitudinal coherence of imputed biochemical trajectories.

4 Exploratory Data Analysis (EDA)

4.1 Data Overview and Summary Statistics

The dataset contains a total of 982 clinical observations spanning across 193 unique patients. Each patient has an average of approximately 5.09 visits, with some longitudinal profiles extending up to 25 separate clinical encounters, providing a valid statistical frame for further analysis.

4.2 Age Demographics and Visit Distribution

The demographic profile of the cohort reveals a mature patient population, with the majority of individuals falling within the 40–60 year age bracket. As illustrated in Figure 1, the age distribution is relatively consistent across both genders, though there is a

Table 2: Summary Statistics of the ADPKD Patient Cohort

Metric	Value
Total Number of Patients	193
Total Observations (Visits)	982
Gender Distribution	115 Males (59.6%), 78 Females (40.4%)
Age Range	16 to 77 years (Mean: 49.5 years)
Avg. Visits per Patient	5.1

slightly higher density of male patients in the middle-age categories. This distribution is characteristic of ADPKD clinical presentations, where symptoms and renal function decline often become most pronounced during the fourth and fifth decades of life.

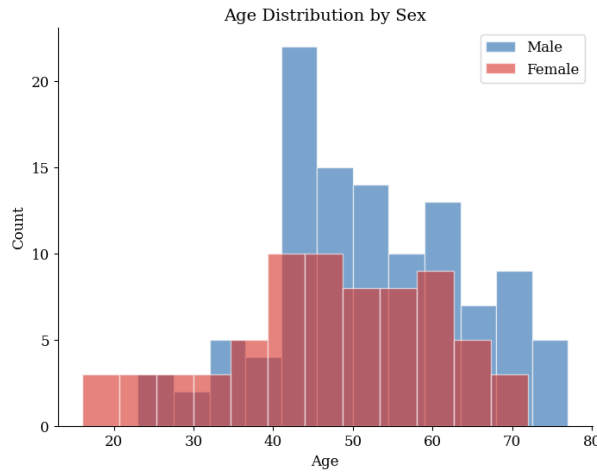


Figure 1: Distribution of Patient Age Categorized by Gender

The longitudinal nature of the dataset is reflected in the visit distribution shown in Figure 2. While a significant portion of the cohort is represented by baseline or early-stage follow-up visits (V1 through V5), a substantial subset of patients demonstrates long-term engagement with the clinical study, extending beyond 10 visits. This variance in visit frequency allows for both cross-sectional analysis of the wider population and deep-dive longitudinal tracking of disease progression in high-frequency patients.

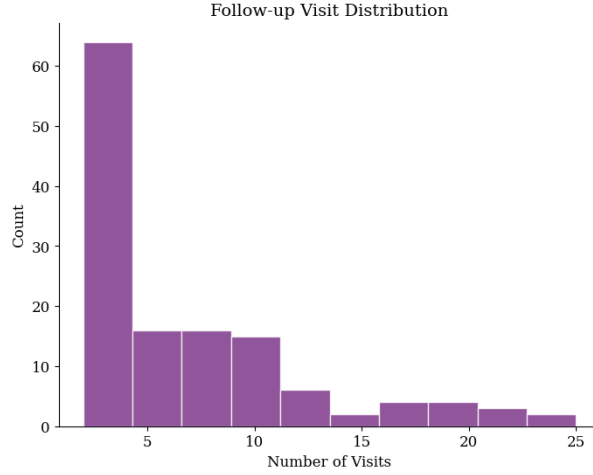


Figure 2: Frequency Distribution of Clinical Visits across the Cohort

The cohort exhibits a broad age range, with a median age of 49 years, indicating that the data captures various stages of ADPKD—from early-onset monitoring to late-stage management in older adults. Notably, there is a high prevalence of Hypertension within the group, which is a hallmark clinical feature of ADPKD. The laboratory data includes essential markers such as Serum Creatinine (Cr), estimated Glomerular Filtration Rate ($eGFR$), and electrolyte levels (K , Na), which will be the focus of further univariate and bivariate analysis to determine their relationship with disease severity.

4.3 Baseline Renal Function and CKD Staging

The baseline renal status of the cohort was evaluated using serum Creatinine levels and the estimated Glomerular Filtration Rate ($eGFR$), both of which are primary indicators of kidney health in ADPKD patients. As shown in Figure 3, the majority of the cohort exhibits baseline Creatinine levels between 0.0 and 5.0 mg/dL, though a significant right-skewed tail indicates a subset of patients presenting with advanced renal impairment at the time of their initial visit.

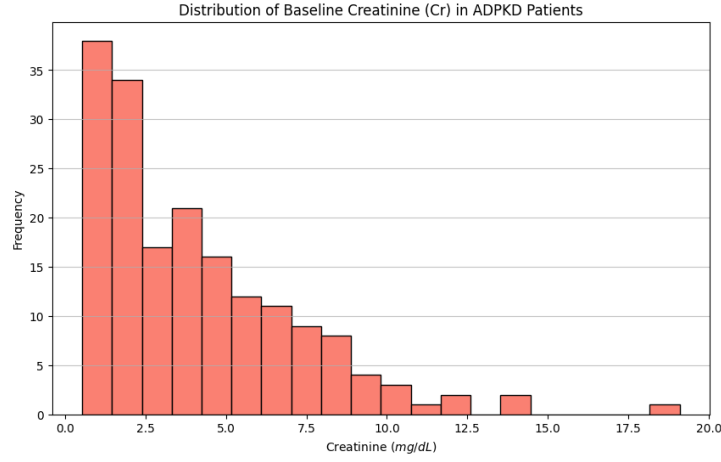


Figure 3: Distribution of Baseline Serum Creatinine Levels

The *eGFR* distribution (Figure 4) further clarifies this impairment, showing a peak in the 0–30 mL/min/1.73m² range. This is also heavily right skewed indicating that the patients were already in a poor clinical state before their first visit to the centre.

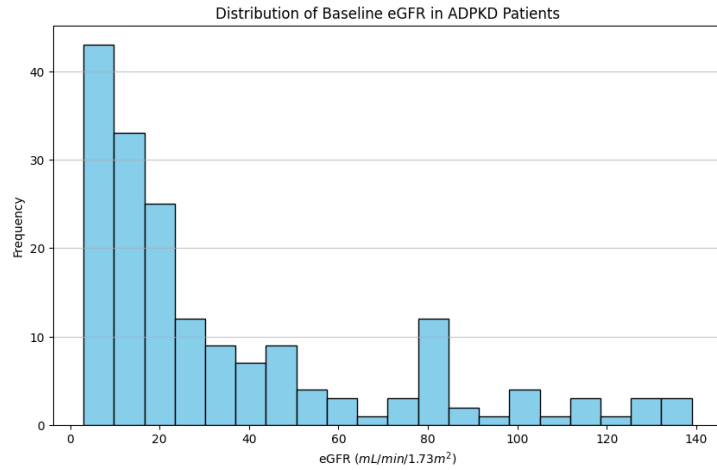


Figure 4: Distribution of Baseline eGFR (mL/min/1.73m²)

To provide a more clinical context, the patients were categorized according to standard Chronic Kidney Disease (CKD) stages (CKD staging followed the KDIGO 2012 classification: G1 (≥ 90), G2 (60–89), G3a (45–59), G3b (30–44), G4 (15–29), and G5 (< 15) mL/min/1.73m².) (Figure 5). Stage 5 represents the largest segment of the population, accounting for over 40% of the patients, representing individuals at high risk of requiring renal replacement therapy. Notably, nearly 15% of the cohort is classified under Stage 3a or Stage 3b. This distribution underscores the severity of the condition within the study group.

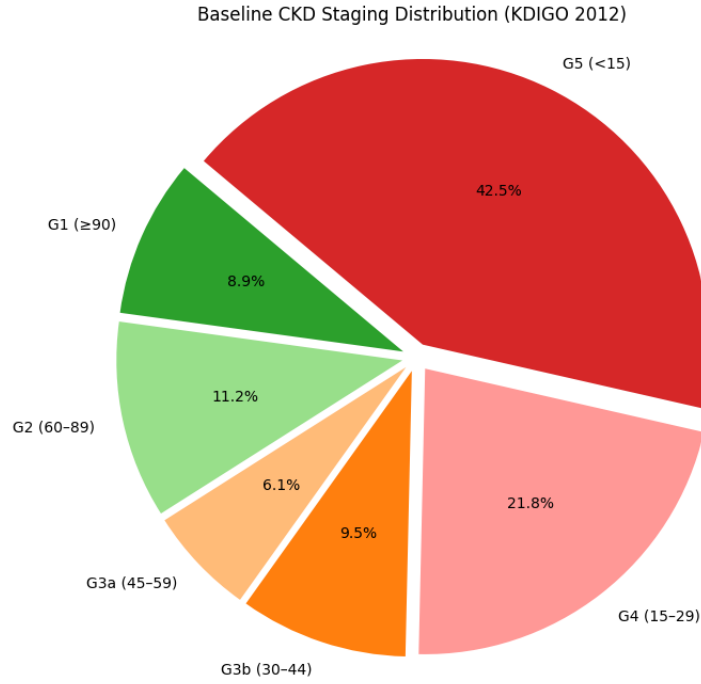


Figure 5: Categorical Distribution of Patients by CKD Stage at Baseline

4.4 Impact of Age and Gender on Renal Function

To further investigate the determinants of renal health within the cohort, the relationship between demographic factors and kidney function markers was analyzed. Figure 6 illustrates the correlation between age and the primary biomarkers, *eGFR* and Serum Creatinine. As expected in a progressive condition like ADPKD, there is a visible downward trend in *eGFR* as patient age increases, accompanied by a corresponding rise in Creatinine levels. This inverse relationship highlights the cumulative nature of cyst growth and parenchymal damage over time, leading to a steady decline in filtration capacity as patients move into their fifth and sixth decades.

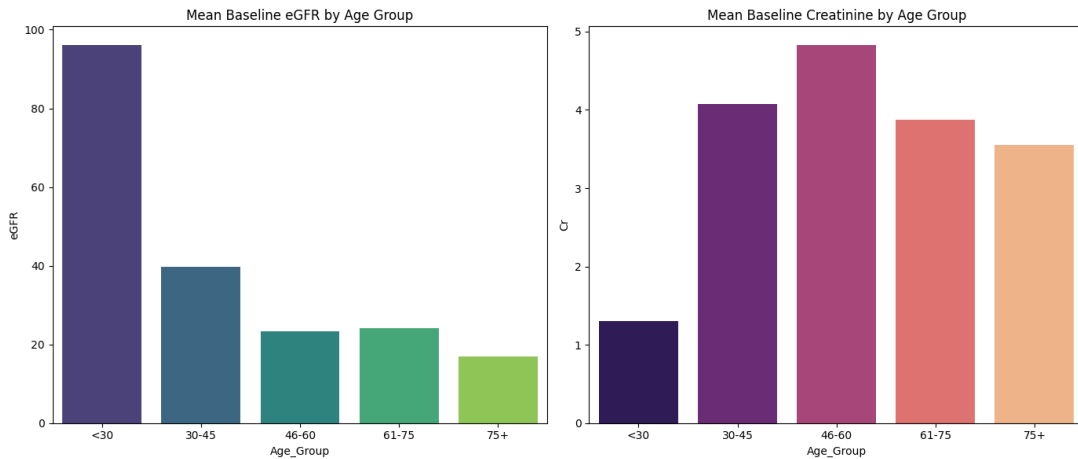


Figure 6: Bar Plots of eGFR and Creatinine vs. Age

Furthermore, the influence of biological sex on renal parameters was assessed, as shown in Figure 7. The boxplots indicate that male patients in this cohort generally present with slightly higher median Creatinine levels and lower median $eGFR$ compared to female patients. While both genders exhibit a wide range of values, the distribution for males shows a greater density of observations in the lower $eGFR$ ranges (< 60 mL/min/1.73m²). This observation aligns with existing clinical literature suggesting that male sex may be a risk factor for more rapid progression in ADPKD, necessitating gender-stratified considerations in long-term prognosis.

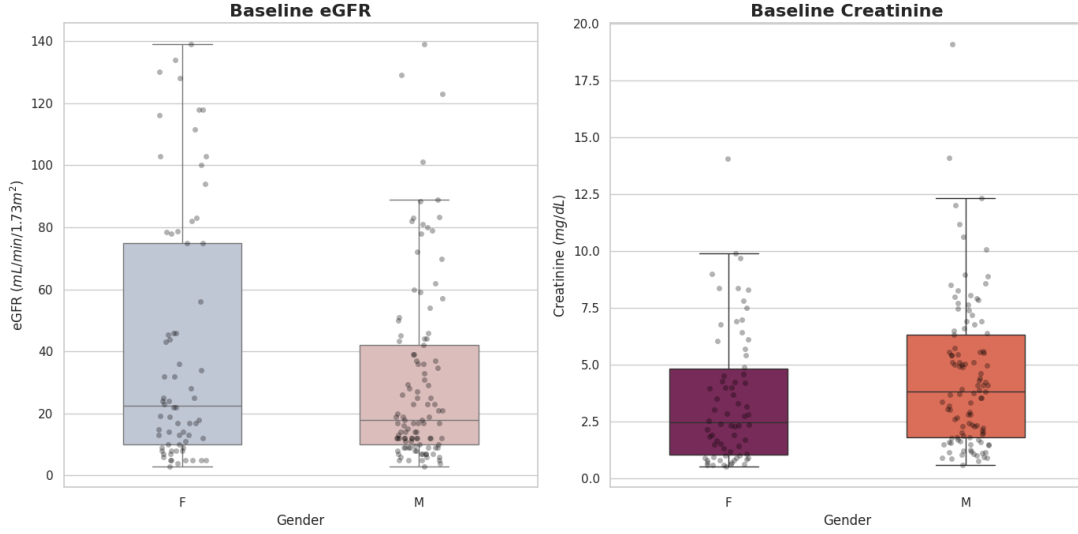


Figure 7: Comparison of eGFR and Creatinine Distributions by Gender

4.5 Distribution of Biochemical Markers by Gender

To assess the systemic impact of ADPKD beyond primary renal markers, the distributions of Potassium (K), Sodium (Na), Hemoglobin (Hb), and Uric Acid were analyzed across genders. This analysis, illustrated across Figures 8 and 9, provides insight into the electrolyte balance and metabolic state of the patient cohort.

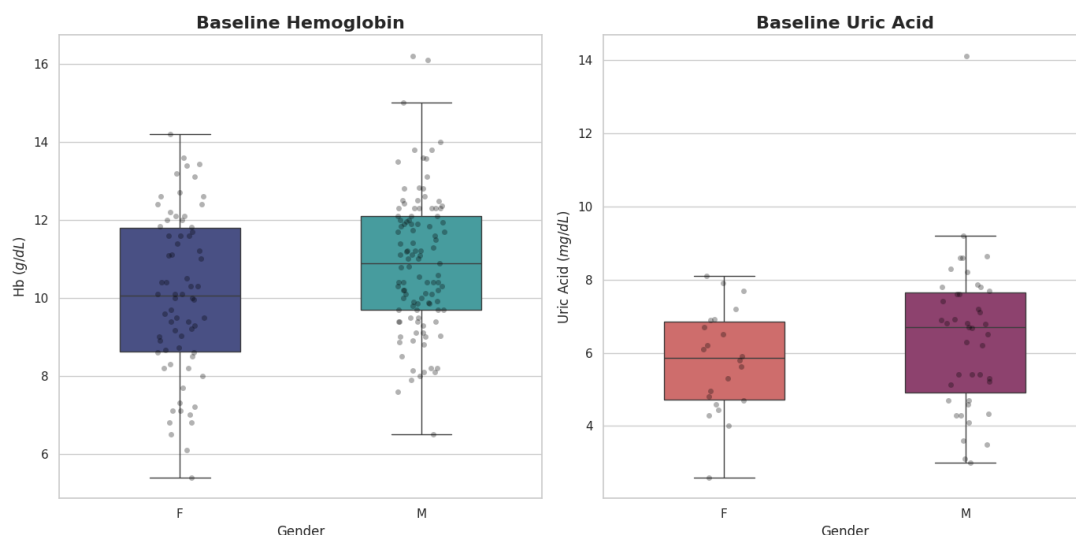


Figure 8: Distribution of Hemoglobin and Uric Acid by Gender

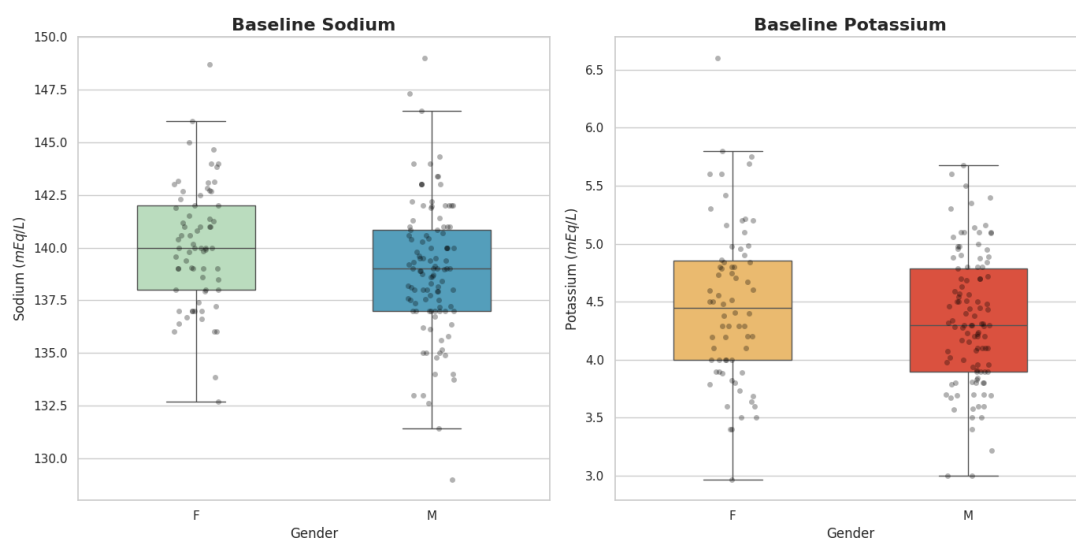


Figure 9: Comparison of Serum Electrolytes (Potassium and Sodium) by Gender

The analysis of these four biochemical markers reveals the following clinical trends:

- **Potassium (K):** As shown in the boxplots, Potassium levels are largely stable, with medians for both sexes residing between 4.0 and 4.5 mEq/L. While the majority of the cohort maintains homeostasis, the presence of several high-end outliers—particularly in the male subgroup—points toward instances of hyperkalemia, a common complication as potassium excretion decreases with declining *eGFR*.
- **Sodium (Na):** Sodium regulation remains remarkably consistent across the entire cohort. Both genders exhibit a tight distribution within the standard physiological range of 135–145 mEq/L. This suggests that despite the structural progression of ADPKD, the kidneys' ability to manage sodium balance remains relatively intact

across most CKD stages represented here.

- **Hemoglobin (*Hb*):** There is a clear sex-based differentiation in Hemoglobin levels. Male patients show a median concentration higher than female patients. The lower quartiles and outliers reaching below 10 g/dL in both groups indicate the onset of renal anemia, likely due to impaired erythropoietin production in patients with more advanced cystic disease.
- **Uric Acid:** Uric Acid levels demonstrate a significant upward shift, especially among male patients whose median exceeds the typical 7.0 mg/dL threshold for hyperuricemia. This elevation is a known metabolic marker for reduced renal clearance and can contribute to further vascular and interstitial damage, potentially accelerating the decline of *eGFR* in these individuals.

4.6 Comorbidity Profile

The clinical profile of the study population ($N = 193$) reveals a high burden of systemic and renal complications, as illustrated in the prevalence analysis.

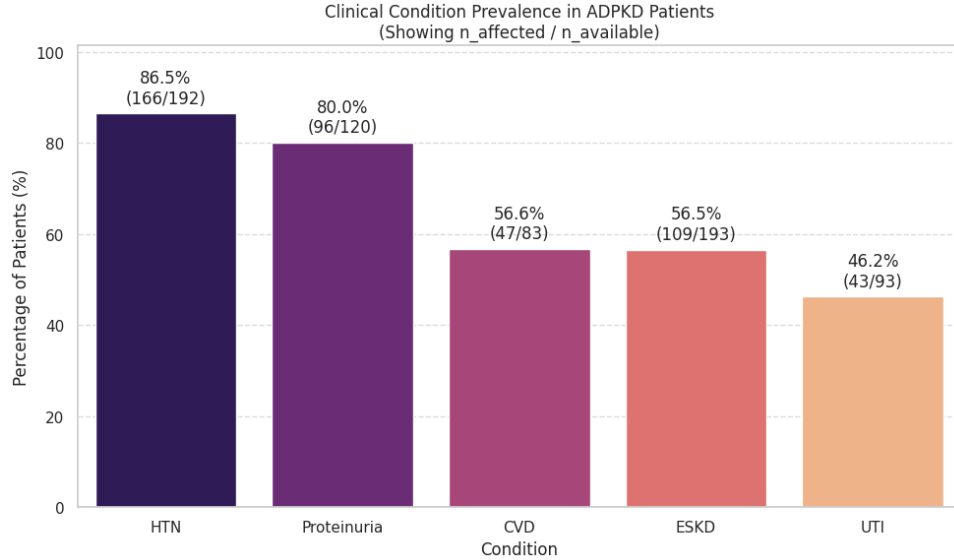


Figure 10: Comorbidity Profile of the cohort

- **Hypertension (86.5%):** Hypertension is the most prevalent comorbidity, affecting the vast majority of the cohort (166/192). This high incidence is consistent with the early activation of the renin-angiotensin-aldosterone system (RAAS) typically observed in ADPKD progression.
- **Proteinuria (80.0%):** A significant proportion (96/120) of patients exhibited proteinuria (defined as $\geq$ trace or > 30 mg/dL). This high prevalence underscores its role as a critical biomarker for renal structural damage and a known predictor of

more rapid disease progression.

- **Cardiovascular Disease (CVD) (56.6%):** Over half of the patients for whom data was available presented with cardiovascular involvement (47/83), including left ventricular hypertrophy (LVH) and valvular abnormalities. This reflects the systemic nature of ADPKD and the cumulative impact of chronic hypertension.
- **End-Stage Kidney Disease (ESKD) (56.5%):** The high rate (109/193) of ESKD (eGFR < 15 mL/min/1.73m² or requirement for renal replacement therapy) suggests a cohort with advanced disease stages, likely reflecting the referral patterns of a tertiary care setting.
- **Urinary Tract Infection (UTI) (46.2%):** Nearly half of the cohort reported instances of UTI (43/93) or recurrent infections. These are common complications in ADPKD, often associated with cyst infections or structural changes that promote urinary stasis and bacterial growth.

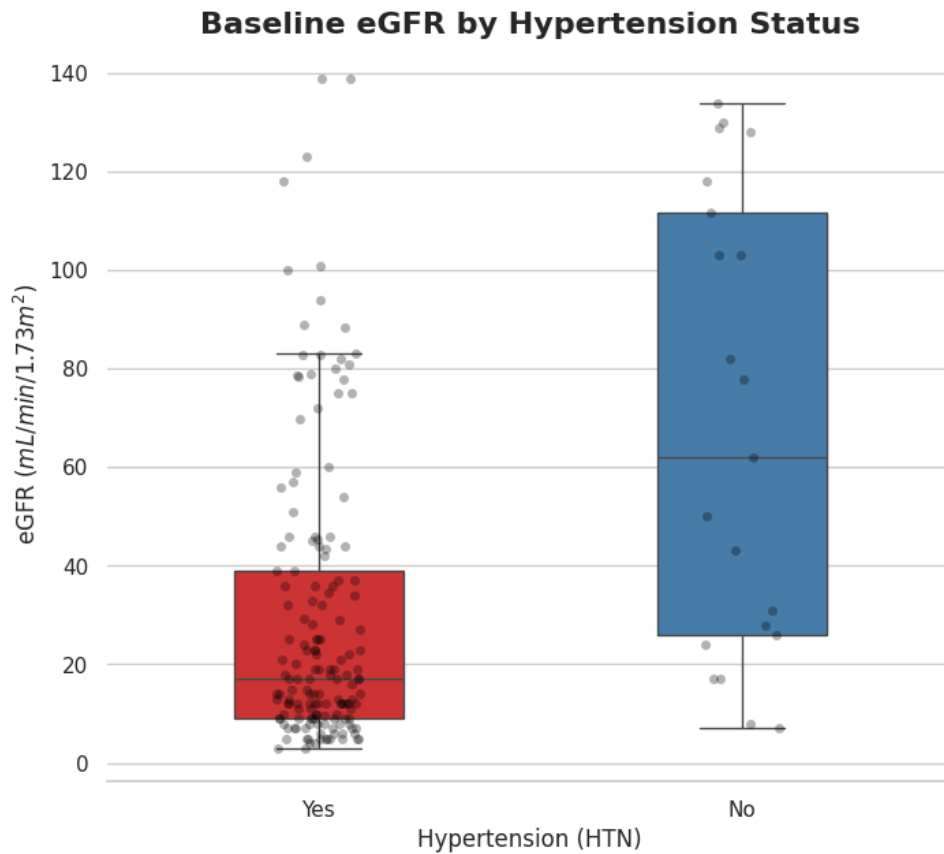


Figure 11: Baseline eGFR across HTN groups

The comparison of baseline renal function reveals a stark disparity in glomerular filtration rates between normotensive and hypertensive patients. In our cohort, patients without hypertension maintain a significantly higher median baseline eGFR, whereas the hyper-

tensive subgroup displays a markedly lower distribution of renal function at the time of initial assessment. This observation is consistent with the known pathophysiology of ADPKD, where the expansion of renal cysts triggers intrarenal ischemia and activates the renin-angiotensin-aldosterone system (RAAS), leading to early-onset secondary hypertension.

The presence of hypertension often serves as a clinical marker for more extensive structural damage within the kidney, as the resulting high systemic pressure further accelerates glomerular sclerosis and interstitial fibrosis. Consequently, the significantly lower baseline eGFR in hypertensive patients underscores the importance of blood pressure as both a symptom of ADPKD progression and a primary driver of further renal decline.

5 Survival Analysis of the Patient Cohort

The present study aimed to evaluate clinically meaningful renal progression outcomes in a retrospective longitudinal cohort of patients with ADPKD. Specifically, the study sought to determine the risk and timing of a composite renal endpoint defined as either a $\geq 40\%$ decline in eGFR from baseline or progression to ESKD. Survival-analysis techniques were employed to estimate progression-free survival over follow-up and to compare progression patterns across clinically relevant subgroups.

In addition to describing progression outcomes, the study further aimed to identify baseline clinical variables independently associated with faster renal progression. Variables including age, gender, hypertension status, haemoglobin level, baseline kidney function, and proteinuria were evaluated as potential predictors of disease progression. Univariable and multivariable Cox proportional hazards regression models were used to quantify the association between these baseline characteristics and the hazard of clinically meaningful renal decline.

Kaplan–Meier survival estimation and log-rank testing were utilized to visualize and compare progression-free survival among patient subgroups, while Cox proportional hazards modelling enabled adjustment for potential confounding variables and estimation of hazard ratios with corresponding confidence intervals. Model adequacy was further assessed using proportional hazards diagnostics and concordance measures.

Through these analyses, the study aims to provide insight into the temporal dynamics of renal progression in ADPKD and to identify clinically relevant determinants of adverse renal outcomes within this cohort. Such findings may contribute to improved risk stratification, patient monitoring, and future development of targeted management strategies for ADPKD progression.

5.1 Study Population

Patients with available longitudinal renal function measurements and follow-up information were included in the analysis. For each patient, baseline clinical variables were extracted from the earliest available visit in the dataset. These baseline variables included:

- Age
- Gender
- Hypertension status (HTN)
- Baseline haemoglobin (Hb)
- Baseline estimated glomerular filtration rate (eGFR)
- Baseline proteinuria status

Patients contributed variable durations of follow-up depending on the number of available visits and progression status. Missing values were handled using variable-specific complete-case analysis, whereby patients were excluded only from analyses involving variables unavailable for that specific analysis.

5.2 Definition of Renal Progression Outcomes

The primary clinically meaningful renal progression endpoint was defined as the occurrence of either:

1. A decline of at least 40% in eGFR from baseline, or
2. Progression to end-stage kidney disease (ESKD)

Baseline eGFR for each patient was defined as the eGFR measurement recorded at the first available visit. A 40% decline event was identified when:

$$\text{eGFR}_{ij} \leq 0.60 \times \text{Baseline eGFR}_i$$

where eGFR_{ij} represents the eGFR of patient i at visit j .

Progression to ESKD was identified using documentation of dialysis initiation or renal transplantation within the clinical records. A composite renal progression endpoint was subsequently defined as the occurrence of either a $\geq 40\%$ decline in eGFR or ESKD.

5.3 Construction of Survival Dataset

A patient-level survival dataset was constructed from the longitudinal records. For each patient, the event time was defined as the first visit at which the composite renal progression

endpoint occurred. Patients who did not experience progression during follow-up were considered censored at their final available visit.

Thus, for each patient:

- Time variable = visit number corresponding to first progression event or last follow-up
- Event indicator = 1 if progression occurred, 0 if censored

This framework enabled the application of time-to-event survival analysis techniques for evaluating progression-free survival.

5.3.1 Estimation of Inter-visit Intervals

The median duration between clinical visits was estimated using a hybrid methodology to maximize the utility of the available longitudinal data. The dataset presented two distinct types of temporal records, which were handled as follows:

For records where specific dates were provided for individual visits (e.g., $V_1, V_2, \dots, V_n$), the duration was calculated precisely. The dates were first standardized to a $DD - MM - YYYY$ format to resolve ambiguities between period and slash delimiters. For each patient j , the intervals Δt were calculated as:

$$\Delta t_i = D_i - D_{i-1} \quad (1)$$

where D_i represents the date of the i -th visit. These calculations capture the real-world stochasticity of patient follow-ups, including acute visits and routine monitoring.

A significant portion of the dataset contained longitudinal summaries in the form of date ranges (e.g., "10/04/2014 to 15/01/2026") rather than discrete dates for intermediate visits. To include these patients without biasing the results toward zero, an *equal-gap assumption* was applied. If a patient had a total of N visits recorded over a span defined by a start date (T_{start}) and an end date (T_{end}), the inter-visit interval was estimated as:

$$\Delta t_{est} = \frac{T_{end} - T_{start}}{N - 1} \quad (2)$$

For each such patient, $N - 1$ identical intervals of length Δt_{est} were added to the aggregate pool of durations.

The final dataset combined $n = 213$ explicit intervals and $n = 500$ estimated intervals, resulting in a total of 713 data points. The *median* was selected as the primary descriptive statistic rather than the mean to ensure robustness against outliers, such as exceptionally long gaps in care (exceeding 3 years) or high-frequency acute visit clusters.

The resulting median inter-visit time was approximately **90 days**, suggesting a typical

clinical follow-up frequency of once per quarter.

5.4 Kaplan-Meier Survival Analysis

Kaplan–Meier survival analysis was performed to estimate progression-free survival probabilities over follow-up and to compare survival patterns across clinically relevant subgroups. Survival curves were interpreted with respect to the composite progression endpoint consisting of either a $\geq 40\%$ decline in eGFR or progression to ESKD. Survival curves were generated for clinically relevant subgroup comparisons including:

- Age groups
- Gender
- Hypertension status
- Baseline CKD stage
- haemoglobin groups
- Proteinuria severity groups

The Kaplan–Meier estimator was used to estimate the survival function:

$$S(t) = P(T > t)$$

where T represents the time to clinically meaningful renal progression.

Differences between survival curves were assessed using log-rank tests. For comparisons involving more than two groups, multivariate log-rank testing was employed.

5.4.1 Overall Risk and Timing of Clinically Meaningful Renal Progression

Kaplan–Meier survival estimation was additionally performed for the overall cohort in order to quantify the cumulative risk and timing of clinically meaningful renal progression, defined as either a $\geq 40\%$ decline in eGFR from baseline or progression to end-stage kidney disease (ESKD).

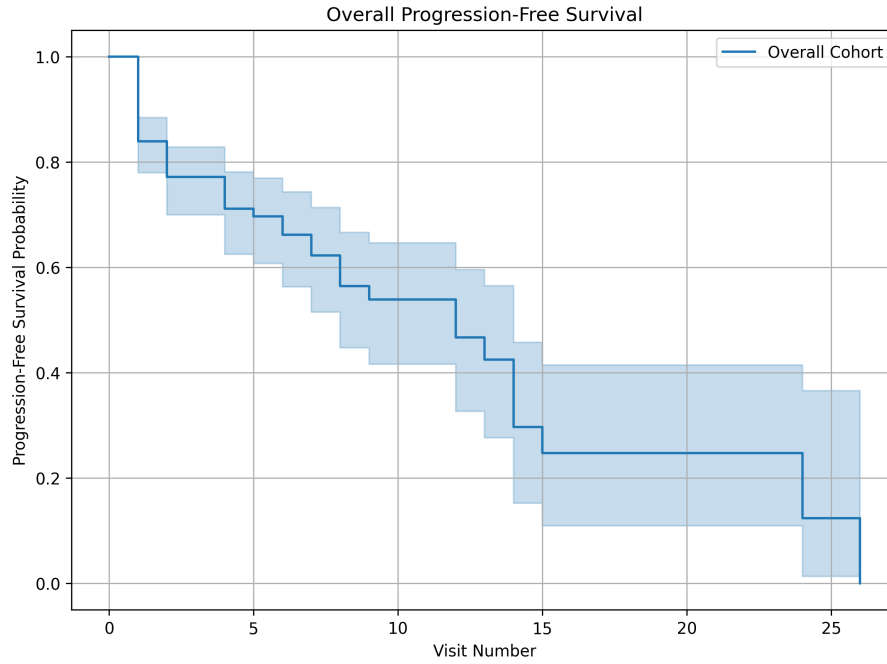


Figure 12: Overall Kaplan–Meier progression-free survival curve for the composite renal progression endpoint.

The Kaplan–Meier survival curve demonstrated progressive decline in progression-free survival probability over longitudinal follow-up, indicating gradual accumulation of clinically meaningful renal progression events within the cohort.

Median progression-free survival for the cohort was approximately 12 follow-up visits, indicating that nearly half of the patients experienced either a $\geq 40\%$ decline in eGFR or progression to ESKD by this time point.

Estimated progression-free survival probabilities at visits 5, 10, and 15 were 69.7%, 53.9%, and 24.8%, respectively. Correspondingly, cumulative progression risks increased progressively during follow-up, reaching approximately 30.3% by visit 5, 46.1% by visit 10, and 75.2% by visit 15.

5.4.2 Categorization of Variables for Kaplan–Meier Analysis

Kaplan–Meier survival analysis requires comparison of survival distributions across discrete patient groups. Therefore, continuous clinical variables were categorized into clinically interpretable groups prior to construction of Kaplan–Meier survival curves and log-rank testing. This categorization was performed solely for visualization and subgroup comparison purposes. In contrast, continuous variables were retained in their original continuous form during Cox proportional hazards regression analysis to preserve statistical information and avoid loss of precision.

Age was categorized into two groups using the cohort median age as the threshold, thereby

creating comparatively younger and older patient subgroups for survival comparison.

Baseline kidney function was categorized according to clinically relevant CKD severity groups derived from baseline eGFR values. Patients were classified as:

- Early CKD: baseline eGFR ≥ 60 mL/min/1.73m²
- Moderate CKD: baseline eGFR between 30 and 59 mL/min/1.73m²
- Advanced CKD: baseline eGFR < 30 mL/min/1.73m²

This grouping enabled clinically meaningful visualization of progression-free survival according to baseline renal disease severity.

Hemoglobin values were similarly categorized using the cohort median hemoglobin level, resulting in higher and lower hemoglobin groups for Kaplan–Meier analysis.

Categorical clinical variables including gender and hypertension status were analyzed directly using their original binary clinical classifications.

Proteinuria data were recorded in heterogeneous formats including qualitative dipstick values, trace categories, and quantitative urinary protein measurements. A standardized ordinal proteinuria severity scale was therefore constructed to harmonize these measurements.

Proteinuria categories were mapped as follows:

Clinical Description	Ordinal Score
Negative / Nil	0
Trace	0.5
1+	1
2+	2
3+	3
4+ / Nephrotic Range	4

For Kaplan–Meier survival analyses, proteinuria severity was further grouped into clinically interpretable categories:

- None (0) (group size = 38)
- Mild (0.5, 1) (group size = 15)
- Moderate-to-Severe (2, 3, 4) (group size = 8)

This approach facilitated clinically meaningful subgroup survival comparisons while accommodating heterogeneous retrospective proteinuria records. Due to substantial missingness in retrospective proteinuria records, analyses involving proteinuria were restricted to patients with available baseline proteinuria measurements.

5.4.3 Age

Kaplan–Meier curves stratified by age group demonstrated modest separation between younger and older patients over follow-up. Older individuals tended to exhibit slightly poorer progression-free survival; however, the curves showed substantial overlap across much of the observation period.

Log-rank testing demonstrated no statistically significant difference in progression-free survival between age groups (log-rank statistic = 3.26, $p = 0.071$).

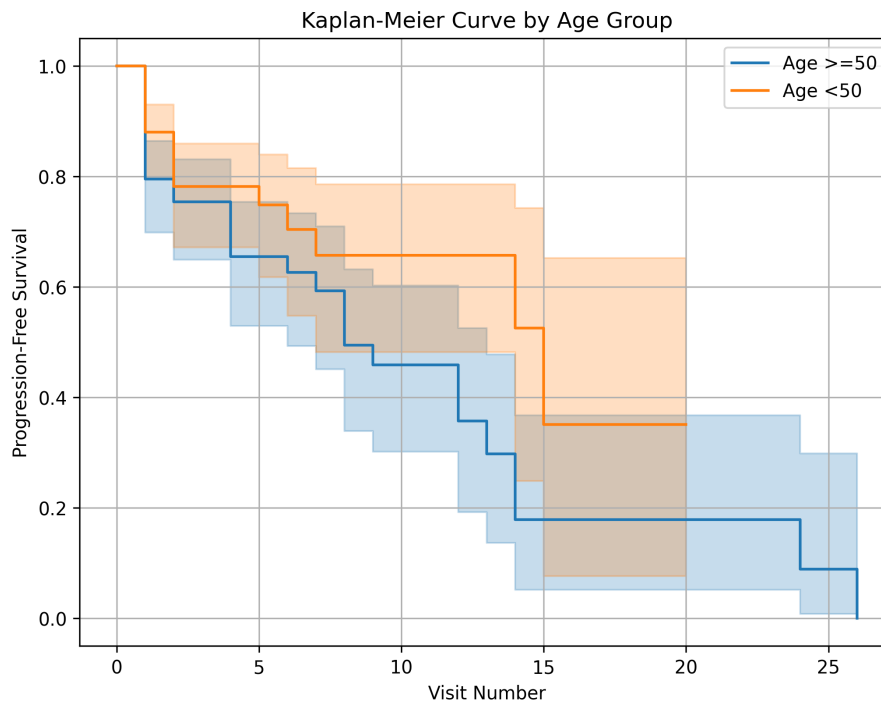


Figure 13: Kaplan–Meier progression-free survival curves stratified by age group.

5.4.4 Gender

Kaplan–Meier survival curves stratified by gender demonstrated relatively similar progression-free survival trajectories between male and female patients throughout follow-up. Although minor differences were observed visually, survival patterns remained broadly comparable between groups.

Log-rank testing did not demonstrate statistically significant differences in renal progression between genders (log-rank statistic = 2.73, $p = 0.098$).

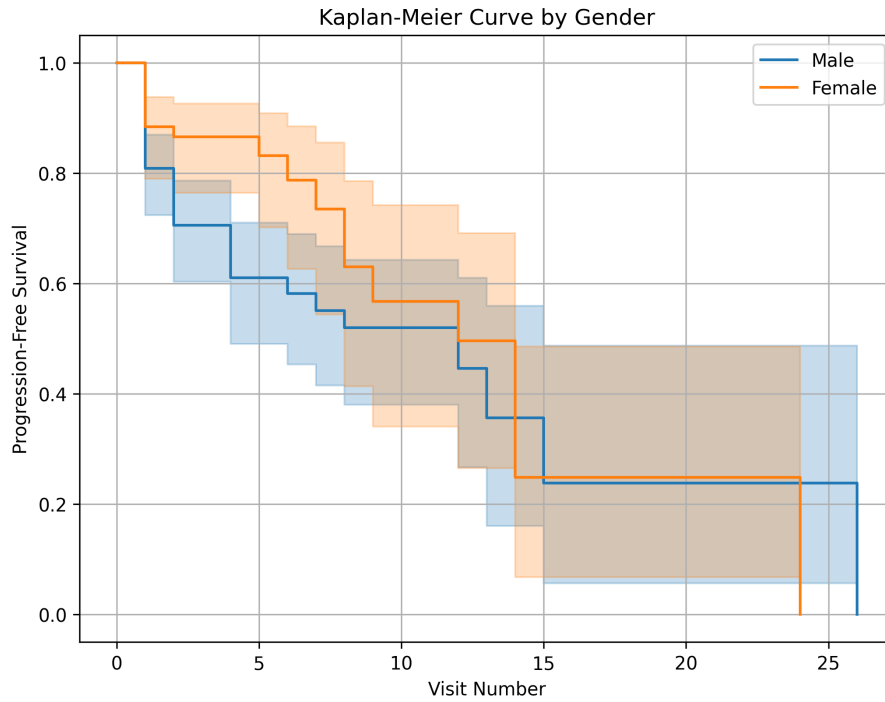


Figure 14: Kaplan–Meier progression-free survival curves stratified by gender.

5.4.5 Hypertension Status

Patients with hypertension demonstrated poorer progression-free survival compared with non-hypertensive individuals, with earlier decline in survival probability observed during follow-up. The separation between curves suggested a potential adverse effect of hypertension on renal progression.

However, log-rank testing did not demonstrate statistically significant differences between hypertensive and non-hypertensive groups (log-rank statistic = 2.57, $p = 0.109$).

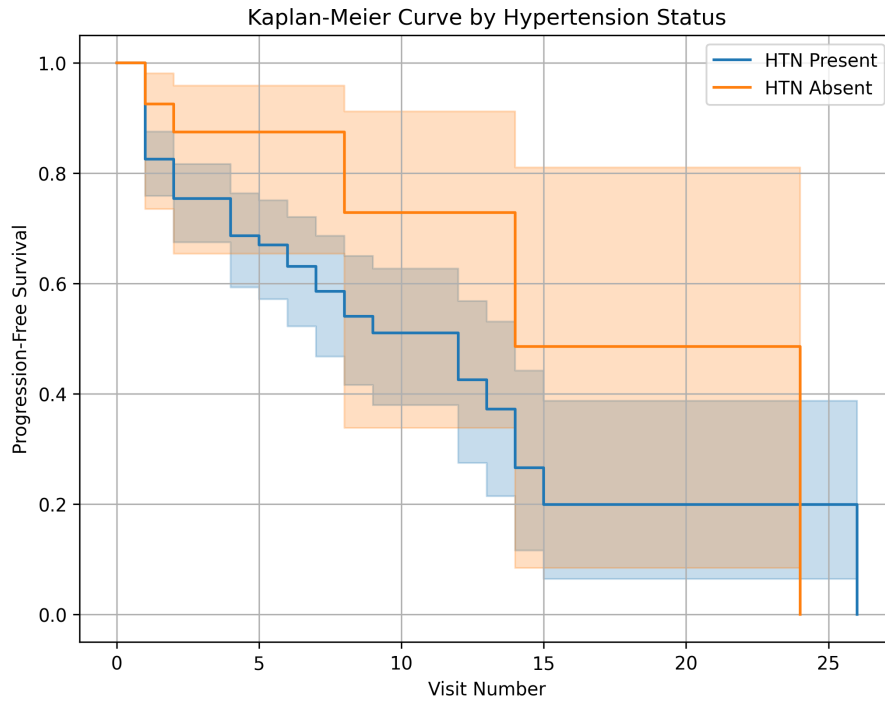


Figure 15: Kaplan–Meier progression-free survival curves stratified by hypertension status.

5.4.6 Baseline Kidney Function

Kaplan–Meier survival curves stratified according to baseline kidney function demonstrated clear separation between CKD severity groups. Patients with advanced baseline renal dysfunction exhibited substantially poorer progression-free survival compared with patients having preserved renal function.

Individuals with lower baseline eGFR experienced earlier and more frequent clinically meaningful renal progression events. Log-rank testing demonstrated statistically significant differences in progression-free survival across CKD severity groups (log-rank statistic = 9.31, $p = 0.0095$), indicating strong unadjusted association between baseline renal function and renal progression.

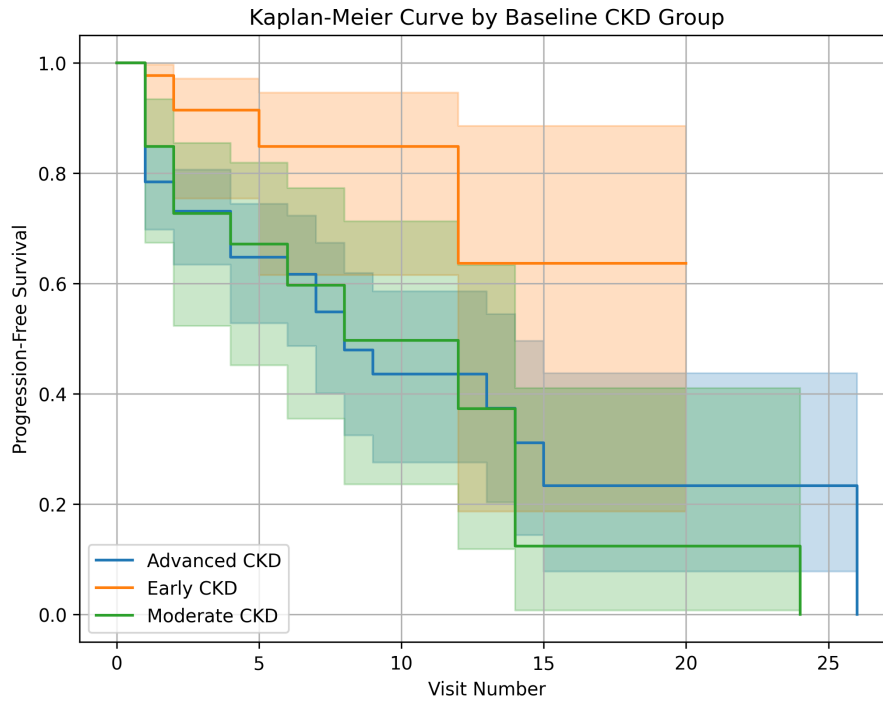


Figure 16: Kaplan–Meier progression-free survival curves stratified by baseline CKD category.

5.4.7 Proteinuria

Kaplan–Meier analysis of proteinuria severity demonstrated progressively poorer survival patterns with increasing proteinuria burden. Patients without proteinuria showed the most favorable progression-free survival, whereas moderate-to-severe proteinuria was associated with earlier progression events.

The observed survival patterns suggested a dose-response relationship between proteinuria severity and renal progression risk. Log-rank testing demonstrated borderline statistical significance across proteinuria severity groups (log-rank statistic = 5.76, $p = 0.056$), suggesting a possible association between increasing proteinuria burden and faster renal progression.

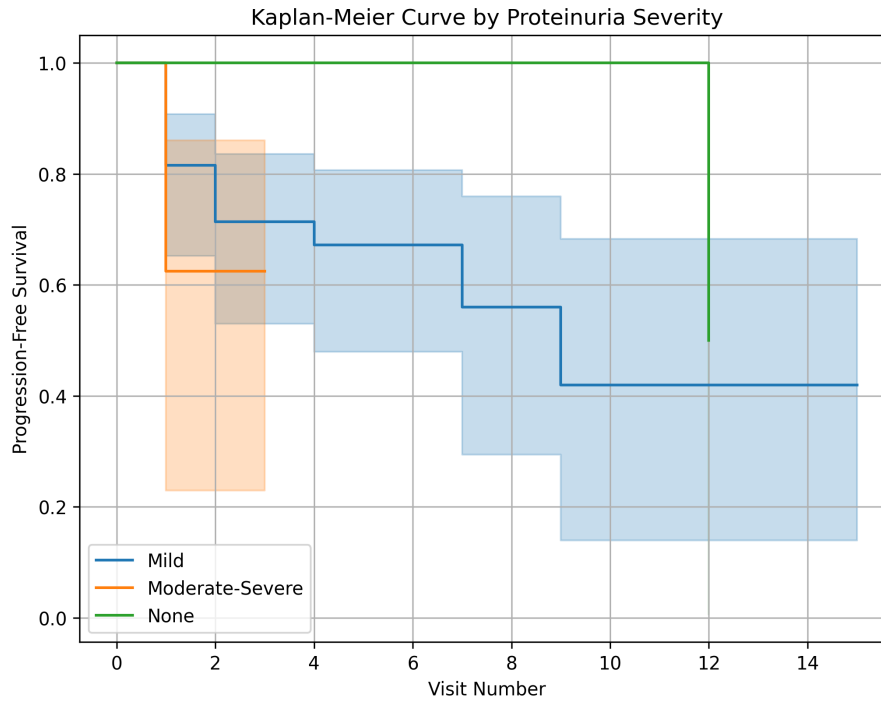


Figure 17: Kaplan–Meier progression-free survival curves stratified by proteinuria severity.

5.4.8 Haemoglobin

Kaplan–Meier survival analysis stratified by haemoglobin level demonstrated poorer progression-free survival among patients with lower baseline haemoglobin concentrations. Individuals with relatively preserved haemoglobin levels tended to maintain progression-free survival for longer durations during follow-up.

Although lower haemoglobin levels visually appeared associated with poorer progression-free survival, log-rank testing did not demonstrate statistically significant differences between haemoglobin groups (log-rank statistic = 2.31, $p = 0.128$).

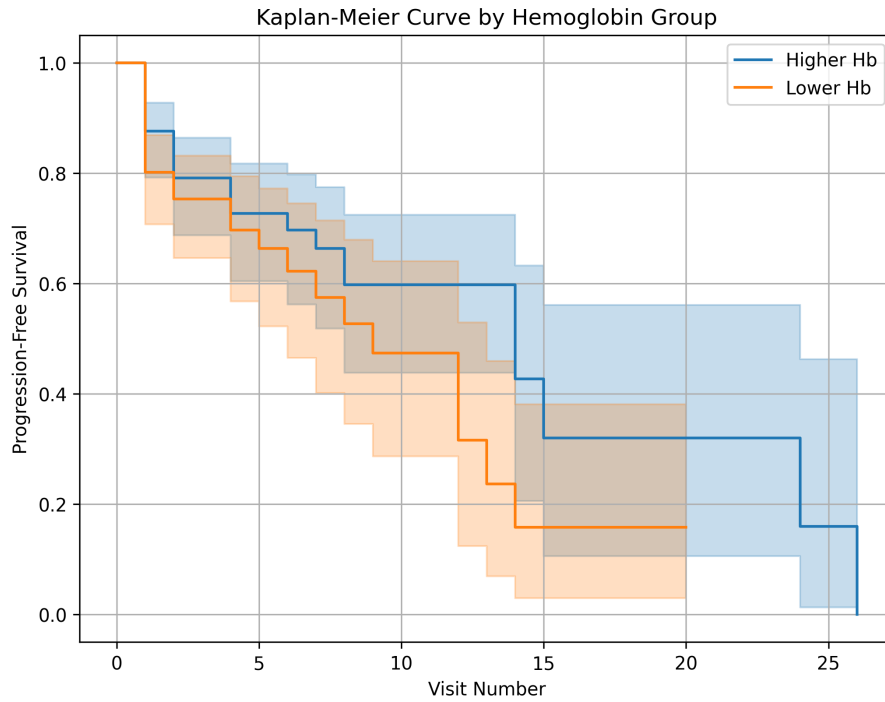


Figure 18: Kaplan–Meier progression-free survival curves stratified by haemoglobin group.

5.4.9 Urinary Tract Infection (UTI)

Kaplan–Meier survival analysis was performed to evaluate progression-free survival according to baseline urinary tract infection (UTI) status. Patients with a history or presence of UTI demonstrated comparatively poorer progression-free survival during follow-up relative to patients without documented UTI.

The Kaplan–Meier curves suggested earlier occurrence of clinically meaningful renal progression events among patients with UTI, although the degree of curve separation varied over follow-up. This observation is clinically plausible because recurrent or severe urinary tract infections in ADPKD may contribute to renal inflammation, cyst infection, and accelerated renal functional deterioration.

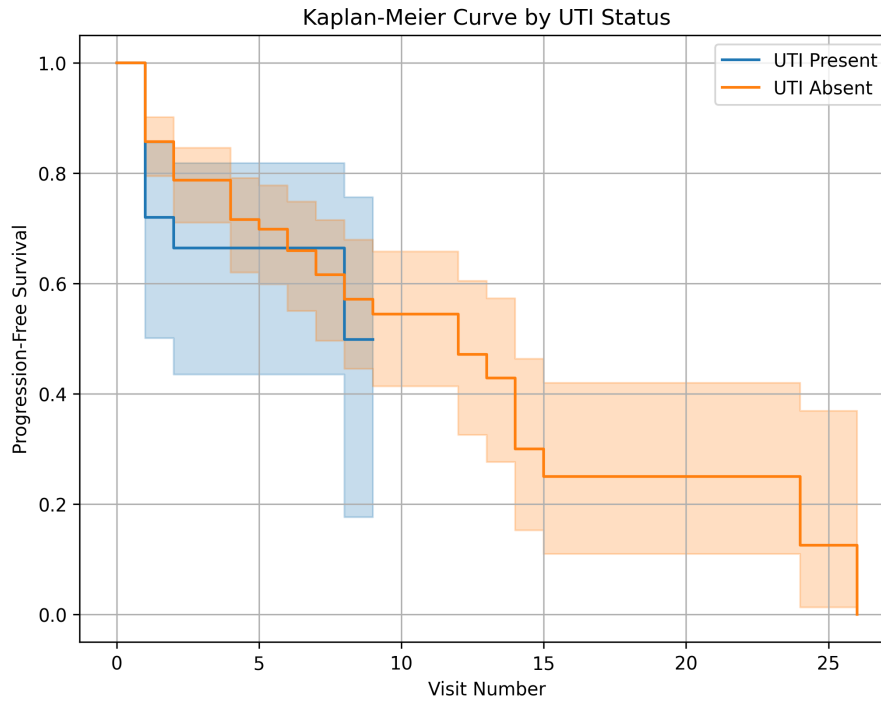


Figure 19: Kaplan–Meier progression-free survival curves stratified by urinary tract infection status.

Log-rank testing demonstrated statistically non-significant differences in progression-free survival between patients with and without UTI (log-rank statistic = 0.95, $p = 0.33$).

5.4.10 Serum Uric Acid

Kaplan–Meier survival analysis was performed to evaluate progression-free survival according to baseline serum uric acid levels. For visualization purposes, patients were categorized into higher and lower uric acid groups using the cohort median serum uric acid value.

The survival curves demonstrated differences in progression-free survival between the two uric acid groups over follow-up. Patients with elevated serum uric acid levels showed altered progression patterns relative to individuals with lower uric acid concentrations, suggesting a potential association between uric acid metabolism and renal disease progression in ADPKD.

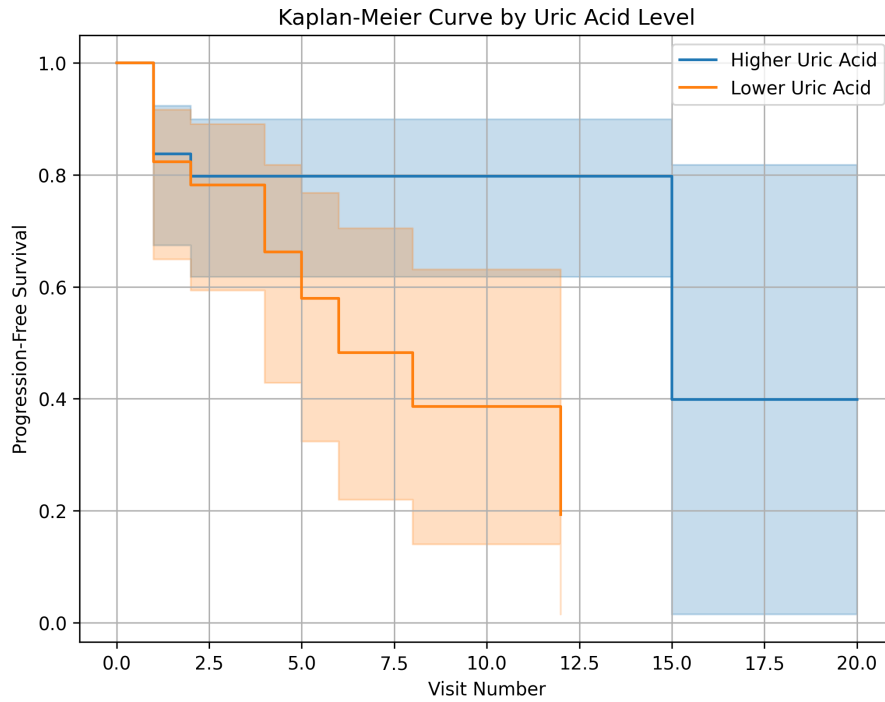


Figure 20: Kaplan–Meier progression-free survival curves stratified by serum uric acid level.

Log-rank testing did not demonstrated statistically significant differences in progression-free survival between the higher and lower uric acid groups (log-rank statistic = 2.275, $p = 0.131$).

5.4.11 Cardiovascular Disease (CVD)

Kaplan–Meier survival analysis was performed to evaluate progression-free survival according to baseline cardiovascular disease (CVD) status. Patients with documented cardiovascular disease demonstrated comparatively poorer progression-free survival during follow-up relative to patients without baseline cardiovascular disease.

The survival curves suggested earlier occurrence of clinically meaningful renal progression events among individuals with cardiovascular comorbidity, indicating a possible relationship between systemic cardiovascular disease burden and adverse renal outcomes in ADPKD.

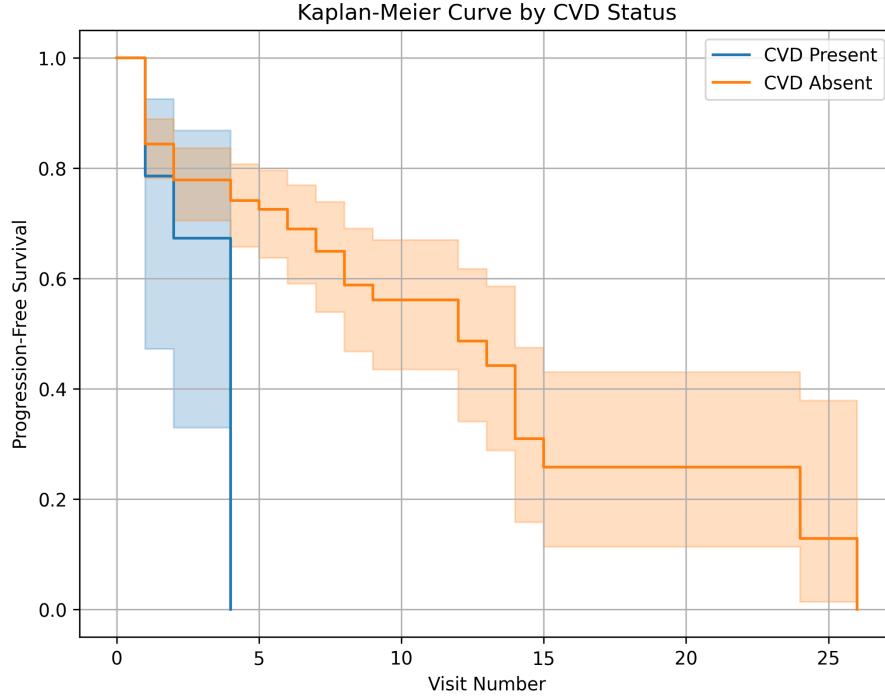


Figure 21: Kaplan–Meier progression-free survival curves stratified by cardiovascular disease status.

Log-rank testing demonstrated no significant differences in progression-free survival between patients with and without cardiovascular disease (log-rank statistic = 3.828, $p = 0.504$).

5.5 Cox Proportional Hazards Regression

To evaluate determinants of renal progression, Cox proportional hazards regression models were fitted using the composite progression endpoint as the outcome variable.

Initially, univariable Cox regression analyses were performed separately for each baseline variable to assess their individual association with progression risk. Continuous variables including age, haemoglobin, and baseline eGFR were analyzed in their continuous form without categorization.

Subsequently, multivariable Cox proportional hazards models were constructed to identify variables independently associated with progression after adjustment for potential confounders. Hazard ratios (HRs), corresponding 95% confidence intervals (CIs), and p-values were reported.

The Cox proportional hazards model is given by:

$$h(t|X) = h_0(t) \exp(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p)$$

where:

- $h(t|X)$ denotes the hazard at time t conditional on covariates X
- $h_0(t)$ denotes the baseline hazard
- β_i represents the regression coefficient associated with covariate X_i

5.5.1 Univariable Cox Proportional Hazards Analysis

Univariable Cox proportional hazards regression analyses were performed to evaluate the independent association of each baseline variable with clinically meaningful renal progression.

Continuous variables including age, baseline eGFR, and haemoglobin were analysed in their continuous form without categorization. Hazard ratios (HRs), corresponding 95% confidence intervals (CIs), and p-values are summarized in Table 3.

Table 3: Univariable Cox proportional hazards regression analysis for clinically meaningful renal progression.

Variable	Hazard Ratio (HR)	95% Confidence Interval	p-value
Age	1.024	1.002 – 1.045	0.029
Baseline eGFR	0.986	0.977 – 0.996	0.005
Proteinuria Severity	2.261	1.217 – 4.203	0.010
Gender (Male vs Female)	1.548	0.913 – 2.624	0.105
Hypertension	1.960	0.838 – 4.587	0.121
Haemoglobin	0.857	0.746 – 0.984	0.029
UTI	1.460	0.710 – 2.979	0.300
Uric acid	0.772	0.570 – 1.032	0.070
CVD	2.309	0.969 – 5.467	0.062

Lower baseline eGFR demonstrated significant association with increased progression hazard in univariable analysis (HR 0.986, 95% CI 0.977–0.996, $p = 0.005$), indicating that poorer baseline kidney function was associated with faster clinically meaningful renal progression. Proteinuria severity additionally demonstrated strong positive association with progression risk (HR 2.261, 95% CI 1.217–4.203, $p = 0.010$), suggesting substantially increased hazard with increasing proteinuria burden.

Increasing age was similarly associated with higher progression hazard (HR 1.024, 95% CI 1.002–1.045, $p = 0.029$). In contrast, higher baseline hemoglobin levels demonstrated a protective association with progression (HR 0.857, 95% CI 0.746–0.984, $p = 0.029$), indicating lower progression risk among patients with preserved hemoglobin concentrations.

Gender and hypertension demonstrated positive but statistically non-significant associations with progression in univariable analysis.

5.5.2 Multivariable Cox Proportional Hazards Analysis

To identify variables independently associated with renal progression after adjustment for potential confounders, multivariable Cox proportional hazards regression modelling was performed.

5.5.3 Initial Multivariable Model Including haemoglobin

The initial multivariable model included:

- Age
- Gender
- Baseline eGFR
- haemoglobin

(Proteinuria and Uric acid excluded for small sample size and CVD is excluded as only 14 patients were CVD)

The results are summarized in Table 4.

Table 4: Initial multivariable Cox proportional hazards regression model including hemoglobin.

Variable	Hazard Ratio (HR)	95% Confidence Interval	p-value
Age	1.011	0.988 – 1.035	0.363
Gender (Male vs Female)	1.564	0.879 – 2.785	0.129
Baseline eGFR	0.991	0.980 – 1.003	0.137
Hemoglobin	0.896	0.750 – 1.071	0.227

Following multivariable adjustment, none of the included variables remained statistically significant. Although the directions of association remained clinically plausible, the

magnitude of statistical evidence attenuated compared with the corresponding univariable analyses.

This reduction in significance likely reflects overlapping prognostic information between baseline kidney function and hemoglobin level. Clinically, lower hemoglobin concentrations are commonly associated with worsening renal dysfunction and advanced CKD severity. Consequently, simultaneous inclusion of hemoglobin and baseline eGFR within the same multivariable model may have introduced collinearity (0.5) and inflated coefficient standard errors, thereby reducing statistical significance despite persistent biologically meaningful trends.

5.5.4 Modified Multivariable Model Excluding Hemoglobin

To reduce potential collinearity between hemoglobin and baseline kidney function, a modified multivariable Cox proportional hazards regression model was subsequently constructed by excluding hemoglobin and incorporating hypertension status alongside age, gender, and baseline eGFR.

Table 5: Modified multivariable Cox proportional hazards regression model excluding hemoglobin.

Variable	Hazard Ratio (HR)	95% Confidence Interval	p-value
Age	1.011	0.987 – 1.035	0.374
Baseline eGFR	0.988	0.978 – 0.999	0.028
Gender (Male vs Female)	1.327	0.764 – 2.308	0.315
Hypertension	1.214	0.484 – 3.046	0.679

In the modified multivariable model, baseline eGFR regained statistical significance after exclusion of hemoglobin (HR 0.988, 95% CI 0.978–0.999, $p = 0.028$), indicating that lower baseline kidney function independently predicted faster clinically meaningful renal progression. The direction of association remained biologically consistent, with reduced baseline eGFR associated with increased hazard of progression.

Age, gender, and hypertension demonstrated non-significant adjusted associations in the final model, although the directions of the estimated hazard ratios remained clinically plausible.

5.5.5 Model Performance

The predictive performance of the multivariable Cox proportional hazards model was assessed using the concordance index (C-index). The concordance index represents the probability that, for a randomly selected pair of patients, the model correctly predicts which patient experiences progression earlier during follow-up. The final model demonstrated a concordance index of 0.650, indicating moderate discriminatory ability for predicting clinically meaningful renal progression.

5.5.6 Residual Analysis Across Proteinuria Severity Groups

To further explore whether proteinuria severity contributed additional unexplained variation beyond the fitted multivariable Cox proportional hazards model, Martingale residuals from the final multivariable model (including age, gender, baseline eGFR, and hypertension status) were examined across proteinuria severity groups.

Patients with available proteinuria measurements were categorized into three clinically interpretable groups:

- None (0) (group size = 38)
- Mild (0.5, 1) (group size = 15)
- Moderate-to-Severe (2, 3, 4) (group size = 8)

Martingale residual distributions for these groups are illustrated in Figure 22.

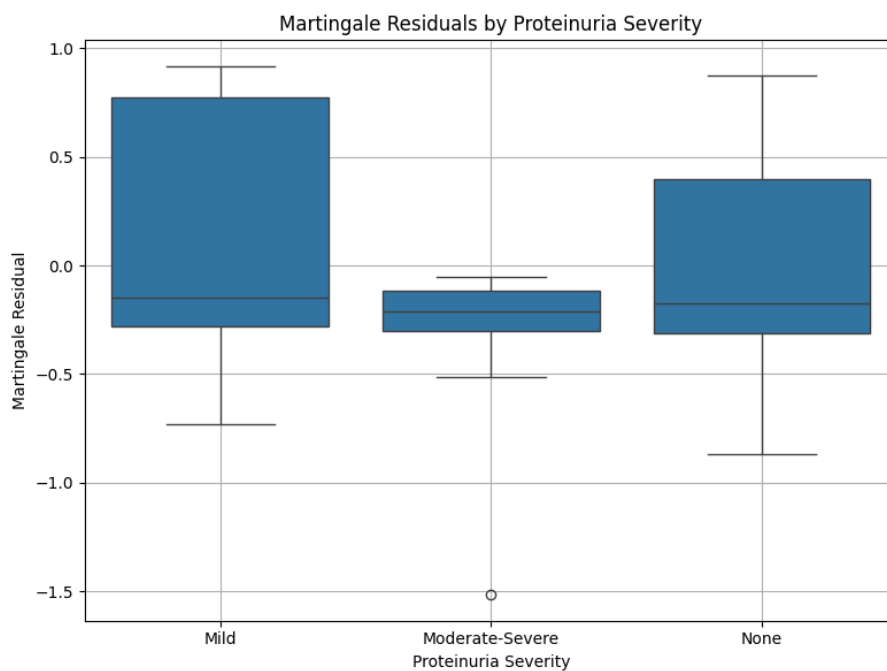


Figure 22: Distribution of Martingale residuals across proteinuria severity groups.

Because residual distributions were non-normal and group sizes were unequal, particularly for the moderate-to-severe proteinuria group, the Kruskal–Wallis nonparametric test was used to compare residual distributions across groups.

The Kruskal–Wallis test demonstrated no statistically significant difference in residual distributions between proteinuria severity categories (Kruskal–Wallis statistic = 1.626, $p = 0.444$).

These findings suggest that substantial progression-related variation associated with proteinuria may already be captured by baseline renal function and other clinical covariates included within the multivariable Cox model. However, interpretation should be performed cautiously because of limited sample size and substantial missingness within retrospective proteinuria measurements.

5.6 Discussion

The present study evaluated clinically meaningful renal progression outcomes in a longitudinal cohort of patients with Autosomal Dominant Polycystic Kidney Disease (ADPKD), with particular emphasis on the timing and risk of progression defined by either a $\geq 40\%$ decline in eGFR or progression to end-stage kidney disease (ESKD). In addition, the study examined baseline clinical variables associated with progression risk using survival-analysis techniques and multivariable Cox proportional hazards modelling.

The analyses demonstrated that baseline renal function was a major determinant of progression risk within the cohort. Kaplan–Meier survival analyses revealed substantially poorer progression-free survival among patients with lower baseline eGFR and more advanced CKD categories. Log-rank testing demonstrated statistically significant differences in survival distributions across baseline CKD groups, indicating strong unadjusted association between impaired baseline kidney function and adverse renal outcomes.

These findings are biologically plausible and consistent with the known pathophysiology of ADPKD. Progressive cyst enlargement and nephron loss ultimately lead to irreversible decline in glomerular filtration rate, and patients presenting with lower baseline eGFR likely represent individuals with more advanced structural kidney disease and reduced renal reserve. Consequently, such patients would be expected to experience clinically meaningful progression events earlier during follow-up.

Univariable Cox regression analyses further supported the importance of baseline kidney function, demonstrating significant association between lower baseline eGFR and increased progression hazard. Increasing age additionally demonstrated significant positive association with renal progression risk, while higher hemoglobin levels were associated with lower progression hazard. Proteinuria severity similarly demonstrated strong association with

adverse progression outcomes.

The observed relationship between proteinuria and progression is clinically important. Although proteinuria is not universally prominent in ADPKD compared with other glomerular disorders, increasing proteinuria burden may reflect more advanced renal parenchymal injury, glomerular hyperfiltration damage, and progressive nephron loss. Kaplan–Meier analyses demonstrated progressively poorer survival patterns with increasing proteinuria severity, suggesting a possible dose-response relationship between proteinuria burden and renal progression.

However, interpretation of proteinuria findings should be performed cautiously because of substantial missingness in retrospective urinary protein measurements. Proteinuria analyses were therefore restricted to a smaller subset of patients, resulting in wider confidence intervals and less stable survival estimates during later follow-up periods.

In multivariable Cox regression analysis, the initial model incorporating age, gender, baseline eGFR, and hemoglobin demonstrated attenuation of statistical significance for all included predictors. This reduction in significance likely reflects overlapping prognostic information between baseline kidney function and hemoglobin level. Clinically, anemia commonly accompanies worsening CKD severity, and lower hemoglobin levels may partially function as surrogate markers of advanced renal dysfunction. Consequently, simultaneous inclusion of hemoglobin and baseline eGFR within the same model may have introduced collinearity and inflated coefficient standard errors.

To address this issue, a modified multivariable model excluding hemoglobin and incorporating hypertension status was subsequently constructed. In this adjusted model, baseline eGFR regained statistical significance, indicating that baseline renal function independently predicted clinically meaningful progression after adjustment for demographic and clinical covariates. This finding reinforces the central role of baseline kidney function as a major prognostic determinant in ADPKD progression.

Hypertension demonstrated positive but statistically non-significant association with progression risk after adjustment. Although hypertension is well established as an important contributor to CKD progression in ADPKD, the absence of statistical significance in the present study may reflect limited event counts, modest sample size, heterogeneous follow-up durations, and retrospective data limitations rather than absence of biological effect.

The concordance index of the final multivariable Cox model demonstrated moderate discriminatory performance for predicting progression outcomes. This suggests that the model possessed reasonable ability to differentiate patients experiencing earlier versus later clinically meaningful renal progression during follow-up.

Several limitations of the present study should be acknowledged. First, the retrospective

observational design resulted in heterogeneous follow-up intervals, incomplete laboratory measurements, and substantial missingness for certain variables, particularly proteinuria. Second, visit number rather than exact chronological follow-up duration was used as the survival time scale because of limitations in available longitudinal timestamp information. Third, the number of progression events was relatively limited for extensive multivariable modelling, potentially reducing statistical power and stability of adjusted estimates. Finally, residual confounding from unmeasured clinical or genetic factors cannot be excluded.

Despite these limitations, the study possesses several important strengths. The analysis utilized longitudinal patient-level follow-up data, clinically meaningful renal progression endpoints, and robust survival-analysis methodology including Kaplan–Meier estimation and Cox proportional hazards modelling. Furthermore, the study contributes longitudinal progression data from an Indian ADPKD cohort, a population that remains relatively underrepresented in the published literature.

Overall, the findings suggest that baseline kidney function is a major determinant of clinically meaningful renal progression in ADPKD. Increasing proteinuria burden, advancing age, and lower hemoglobin levels additionally demonstrated adverse progression associations in unadjusted analyses. These findings may assist in identifying higher-risk patients requiring closer monitoring and earlier renoprotective interventions within routine clinical practice.

6 Estimated Glomerular Filtration Rate (eGFR) Decline Slope

Understanding the rate of decline in eGFR among patients with ADPKD is crucial, as it provides a direct measure of disease progression and kidney function deterioration. Quantifying the decline through slope estimates allows for a more precise and dynamic assessment compared to single time-point measurements. This analysis enables the identification of high-risk individuals who may experience rapid progression, thereby supporting early intervention and personalized treatment strategies. Furthermore, examining average slopes across different cohorts helps uncover heterogeneity in disease trajectories, which may be driven by demographic, genetic, or clinical factors. By incorporating multiple covariates into a regression framework, the study provides insights into the relative importance and interaction of these factors in influencing disease progression. Such findings are valuable for improving prognostic models and guiding clinical decision-making. Additionally, this approach contributes to better trial design by identifying meaningful endpoints and relevant patient subgroups. Overall, the analysis enhances our understanding of ADPKD progression and supports more effective management of the disease.

6.1 Outline of this section

This project focuses on analyzing data related to Autosomal Dominant Polycystic Kidney Disease (ADPKD), a progressive condition characterized by declining kidney function over time. A key clinical marker in this context is the estimated Glomerular Filtration Rate (eGFR), whose rate of decline reflects disease progression. We aim to quantify this progression by estimating individual decline slopes of eGFR for patients. The slope is visit wise, and we assume that the visits are uniformly spread. These slopes will then be summarized across different cohorts to understand variation in disease trajectories. By comparing average slopes, we can identify patterns associated with demographic and clinical subgroups. The study further incorporates relevant covariates such as age, sex, and clinical measurements. Using these factors, we will construct a regression model with the eGFR decline slope as the response variable. This allows us to evaluate the influence of multiple predictors on disease progression. The analysis provides both descriptive and inferential insights into ADPKD progression. Ultimately, the findings may help improve risk stratification and guide clinical decision-making.

6.2 Data processing

The dataset considered in this study consists of longitudinal observations, where each patient has multiple recorded visits over time. To ensure reliable estimation of disease progression, we restrict our analysis to patients with at least 2 visits. However, we shall see that, for patients with 2 visits, the data becomes unreliable, hence will concentrate primarily on the patients with ≥ 3 visit. For each patient, the rate of decline in kidney function is quantified by defining the slope as the regression coefficient of visit number, with eGFR as the response variable. This provides an individual-specific summary of disease progression over time. The eGFR is modelled as

$$\text{eGFR}_{ij} = \alpha_i + \beta_i \text{Visit}_j + \epsilon_{ij}$$

and the least square estimate $\hat{\beta}_i$ of β_i is treated as the eGFR slope of the patient. In addition to the longitudinal measurements, baseline characteristics such as health parameters and age at the time of the first visit are collected. These baseline covariates are then used to explain variability in the estimated slopes across patients. This framework allows us to connect initial patient profiles with subsequent rates of kidney function decline. By doing so, the analysis captures both within-patient temporal trends and between-patient heterogeneity.

6.3 Exploratory results

We have obtained slopes for 132 patients, who have visited the clinic atleast twice, and the summary statistics are summarised at table 1.

Mean	SD	Median	Min	Max	Range
0.13	13.15	-0.69	-70	108	178

Table 6: Summary statistics of eGFR Slope

We can see that, there are large outliers, both in positive and negative sides. One explanation can be inclusion of patients with very few visits. Hence, for patients with ≥ 3 visits, we obtain slopes for 101 patients, and the summary statistics are are summarised at table 2.

Mean	SD	Median	Min	Max	Range
-0.71	4.24	-0.7	-21.5	14.5	36

Table 7: Summary statistics of eGFR Slope

It is visible that we have much less outliers, and as a result, the variance stabilizes. Now we will explore the slopes for different cohorts, depending upon age group, gender, and baseline eGFR rate, in table 3, and 4.

Gender	observations	Mean	SD	Median	Min	Max	Range
F	34	0.28	4.25	-0.83	-4.04	14.5	18.54
M	67	-1.21	4.18	-0.69	-21.5	8.8	30.3

Table 8: Summary statistics of eGFR Slope by gender

Age Group	Gender	observations	Mean	SD	Median	Min	Max	Range
< 30	F	3	0.41	6.41	-2.49	-4.04	7.76	11.8
< 30	M	1	-11	NA	-11	-11	-11	0
30–50	F	18	1.28	5.23	-0.45	-3.62	14.5	18.12
30–50	M	29	-0.89	4.72	-0.54	-21.5	4.63	26.13
50–70	F	12	-1.16	0.66	-1.15	-2.64	-0.12	2.52
50–70	M	31	-0.90	3.54	-0.69	-12.2	8.8	21
70+	F	1	-0.85	NA	-0.85	-0.85	-0.85	0
70+	M	6	-2.70	2.90	-2.05	-7.8	0.10	7.7

Table 9: Summary statistics of eGFR Slope by age group and gender (for $k \geq 3$ visits)

Baseline eGFR was categorized into clinically relevant groups based on standard CKD staging thresholds. The cutoffs (0–15, 15–30, 30–60, 60–90, > 90 ml/min/1.73 m²) reflect progressively worsening kidney function. These categories align with established clinical guidelines and allow meaningful interpretation of disease severity. Grouping helps capture the nonlinear progression of renal decline across different stages. Additionally, this binning improves interpretability while maintaining sufficient sample size within groups for analysis. The table 10 summarises the findings.

eGFR Group	Gender	observations	Mean	SD	Median	Min	Max	Range
0–15	F	7	-0.57	0.37	-0.70	-1.11	-0.02	1.08
0–15	M	29	0.51	2.02	-0.21	-1.80	6.23	8.03
15–30	F	8	0.12	3.22	-0.86	-1.85	7.99	9.85
15–30	M	15	-0.87	2.95	-1.14	-3.74	8.80	12.54
30–60	F	8	-2.17	1.31	-2.10	-4.04	-0.12	3.92
30–60	M	15	-4.26	6.40	-2.98	-21.5	4.53	26.03
60–90	F	4	3.80	7.98	2.17	-3.62	14.5	18.12
60–90	M	6	-1.50	1.79	-0.84	-3.91	0.54	4.45
> 90	F	7	2.09	5.69	0	-2.54	12.27	14.81
> 90	M	2	-4.74	8.85	-4.74	-11	1.51	12.51

Table 10: Summary statistics of eGFR Slope by eGFR group and gender

6.4 Estimation of eGFR slopes

The rate of decline in kidney function, measured by the eGFR slope, reflects disease progression in patients. Clinical biomarkers such as serum creatinine (CR), sodium (Na), potassium (K), and hemoglobin (Hb), along with demographic factors like age and gender, are known to influence renal function and its deterioration. Regressing eGFR slope on these covariates allows us to quantify their association with disease progression and identify significant predictors of faster or slower decline.

We consider the multiple linear regression model:

$$\text{slope}_i = \beta_0 + \beta_1 \text{CR}_i + \beta_2 \text{Na}_i + \beta_3 \text{K}_i + \beta_4 \text{Hb}_i + \beta_5 \text{Age}_i + \beta_6 \text{Gender}_i + \varepsilon_i, \quad i = 1, \dots, n,$$

where:

- slope_i denotes the estimated eGFR decline rate for the i -th individual,
- $\text{CR}_i, \text{Na}_i, \text{K}_i, \text{Hb}_i$ are clinical measurements at baseline,
- Age_i is the age at baseline,
- Gender_i is an indicator variable (e.g., 1 for male, 0 for female),
- $\varepsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$.

Before the estimation, we want to address the problem of multicollinearity. In this regard, we will obtain the correlation matrix, and explore the possibility of redundancy.

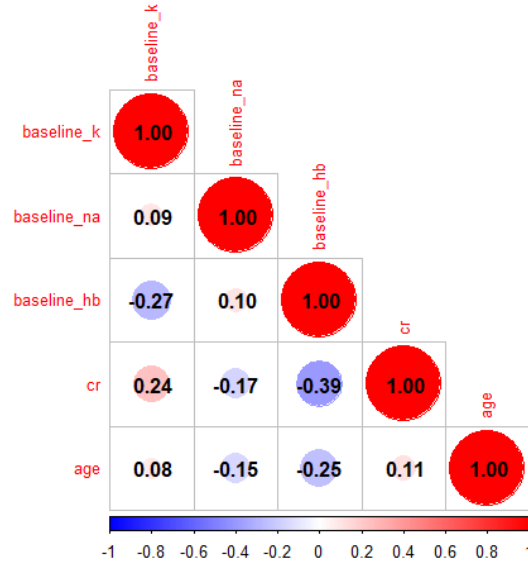


Figure 23: Correlation matrix of the predictor variables

Hence, we can see, the correlations are not much, and hence we can work with these predictors. Note that, eGFR count of a patient is a function of his/her gender, age and serum creatinine level, and hence, is not included as a factor. The summary of the regression is available in table 11.

Variable	Estimate	Std. Error	t value	p-value
Intercept	7.04917	18.30847	0.385	0.7011
serum creatinine	0.45106	0.17693	2.549	0.0124
Hb	0.31580	0.29015	1.088	0.2792
K	-0.19832	0.79081	-0.251	0.8025
Na	-0.06266	0.12849	-0.488	0.6269
Age	-0.03098	0.03810	-0.813	0.4182
gender(M)	-2.37142	1.01150	-2.344	0.0212

Table 11: Linear regression results. Multiple $R^2 = 0.1044$, Adjusted $R^2 = 0.04722$.

Note that, the female gender is considered as the baseline.

The model explains a modest proportion of the variability in the outcome ($R^2 = 0.1044$,

adjusted $R^2 = 0.0472$), suggesting limited overall explanatory power. Among the covariates, serum creatinine (cr) is positively and significantly associated with the slope ($p = 0.0124$), indicating that higher baseline creatinine is linked to a less negative (or slower declining) eGFR slope. It is plausible that in more advanced stages of kidney disease, the rate of further decline in eGFR may attenuate, as physiological limits are approached and progression dynamics become nonlinear. This may partly explain the observed association between higher baseline creatinine and a less steep decline in eGFR. However, such effects can also arise due to regression to the mean or measurement artifacts, and should therefore be interpreted with caution. Gender also shows a statistically significant effect, with males exhibiting a lower slope compared to females ($p = 0.0212$), implying faster decline in kidney function.

Other variables, including baseline hemoglobin, potassium, sodium, and age, do not show statistically significant associations at conventional levels. The overall F-test is not significant ($p = 0.1023$), indicating that the model as a whole may not provide a strong fit to the data. These findings suggest that while certain predictors such as creatinine and gender are relevant, additional variables or alternative model specifications may be required to better explain variation in eGFR decline. Availability of more parameters, or exact gap between the visits, would have made this a more accurate model.

6.5 Exploratory assessment using residual boxplots

Due to the unavailability of hypertension and proteinuria variables in the regression framework, their potential effects were explored through residual diagnostics. Specifically, boxplots of model residuals were examined across different levels of proteinuria and hypertension status. These plots provide a visual assessment of whether systematic variation remains unexplained by the fitted model.

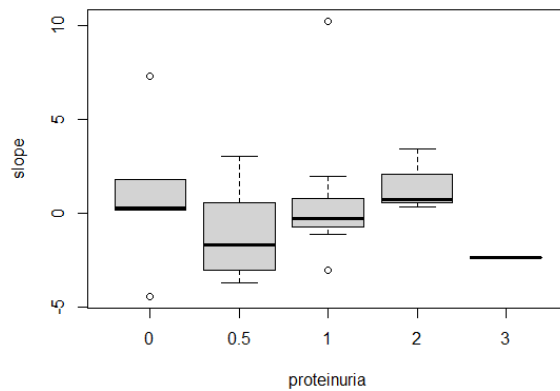


Figure 24: Box plot of residuals with baseline proteinuria levels

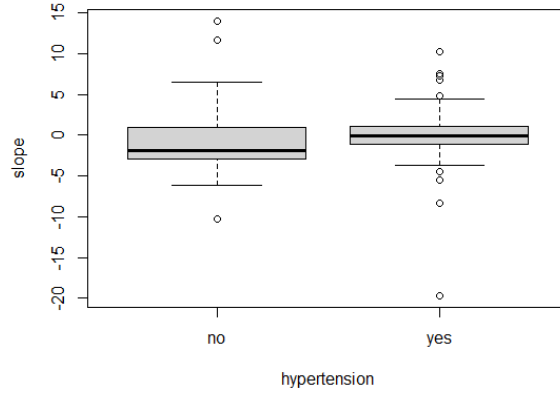


Figure 25: Box plot of residuals with baseline hypertension levels

It is observed that the distribution of residuals varies across groups, with noticeable shifts in central tendency and spread for certain proteinuria levels and hypertensive categories. This suggests that these variables may have an underlying association with the response that is not captured in the current model specification. In particular, heterogeneity in residual dispersion indicates possible omitted variable effects or model misspecification.

While these findings are exploratory and not inferential, they highlight the importance of incorporating such clinical variables in future analyses. Their inclusion may improve model fit and provide a more comprehensive understanding of factors influencing eGFR decline.

6.6 Limitations and further works

This study is subject to several limitations arising primarily from the structure and availability of the data. First, the longitudinal measurements are recorded at irregular visit intervals; however, the analysis assumes approximately uniform spacing between visits when estimating eGFR slopes. This simplification may introduce bias in the estimated rates of decline.

Second, the dataset lacks important clinical and contextual variables, such as precise time gaps between visits, treatment history, and comorbid conditions, all of which may influence disease progression. The absence of these variables limits the explanatory power of the regression model, as reflected in the relatively low R^2 value.

Third, stratification by age, gender, and eGFR groups leads to small sample sizes in several subgroups, resulting in unstable summary statistics and reduced reliability of subgroup comparisons. Additionally, the presence of extreme values in slope estimates suggests potential measurement variability or insufficient observations for certain patients.

Future work can address these limitations by incorporating more detailed and structured data. In particular, using exact timestamps for visits would allow for more accurate estimation of eGFR decline rates. A natural extension is to employ mixed-effects or longitudinal models, which can better account for within-patient variability and irregular observation schedules.

Further improvements can be achieved by including additional clinically relevant covariates, such as treatment information, genetic markers, and comorbidities, to enhance predictive performance.

7 Longitudinal CKD Progression

The analysis set comprised 98 patients contributing 753 longitudinal observations. Because calendar dates were inconsistently recorded for many visits, longitudinal time was defined using visit order within each patient. The first recorded visit for each patient was coded as $t = 0$, the second as $t = 1$, and so forth. This produced a patient-specific longitudinal timescale while preserving the ordering of repeated observations.

Summary statistics indicated substantial heterogeneity in both follow-up intensity and renal function:

Variable	Min	1st Qu.	Median	Mean	Max
eGFR	3	10	21	34.26	139
Baseline proteinuria	0	0	0.5	0.76	3
Visit-based time	0	1	4	5.67	25

The number of visits per patient was highly variable, with a strongly right-skewed distribution (Figure 26). While many patients contributed relatively short follow-up histories, a smaller subset had prolonged longitudinal follow-up extending beyond 20 visits.

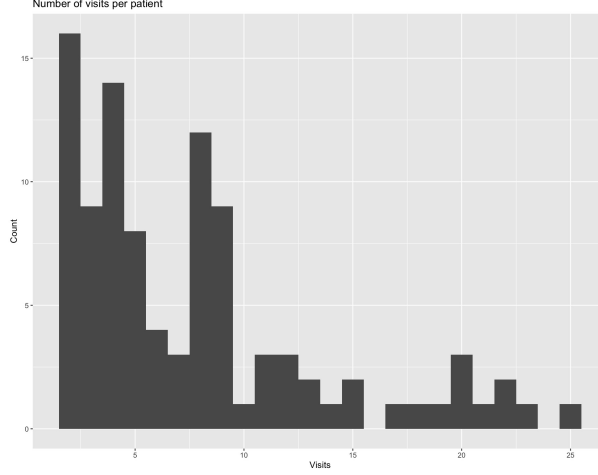


Figure 26: Distribution of longitudinal follow-up intensity measured by number of visits per patient.

Exploratory trajectory plots revealed substantial between-patient heterogeneity in renal progression patterns (Figure 27). Patients differed markedly both in baseline eGFR levels and in longitudinal progression slopes. Some individuals exhibited relatively stable renal trajectories, whereas others showed rapid decline or substantial variability across visits. These plots strongly suggest that repeated measurements within the same patient are correlated and cannot reasonably be treated as independent observations.

Remark: Some of the patients show a sudden jump in their eGFR levels. We suspect that this could be due to some medical intervention such as transplant. We inquired about this from the medical practitioner who had shared this data with us, but received no reply. We have therefore removed those *three* patients from our analysis whose trajectories showed a jump of 60 units or more between two consecutive visits.

The association between baseline proteinuria and baseline renal function was examined next (Figure 28). Higher baseline proteinuria appeared to be associated with lower baseline eGFR, suggesting that proteinuria may reflect more advanced renal impairment at study entry.

To visualize the average longitudinal behavior of kidney function, smoothed trajectories were plotted against visit-based time (Figure 29). The mean trajectory suggested overall decline in renal function over time, although uncertainty increased substantially at later visits because relatively few patients contributed long follow-up histories.

As an initial benchmark analysis, a naive linear regression model was fitted:

$$\begin{aligned} \text{eGFR}_{ij} = & \beta_0 + \beta_1 t_{ij} + \beta_2 \text{HTN}_i + \beta_3 \text{Proteinuria}_i + \beta_4 \text{Baseline eGFR}_{i,c} \\ & + \beta_5 t_{ij} \text{HTN}_i + \beta_6 t_{ij} \text{Proteinuria}_i + \beta_7 t_{ij} \text{Baseline eGFR}_{i,c} + \varepsilon_{ij}, \end{aligned}$$

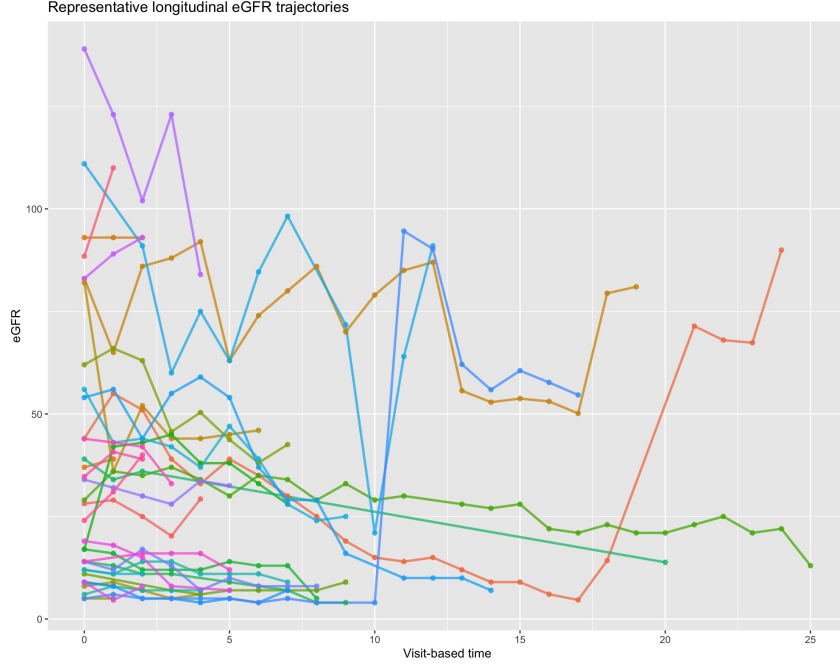


Figure 27: Individual longitudinal eGFR trajectories across visit-based time.

where $\text{Baseline eGFR}_{i,c}$ denotes centered baseline eGFR.

This model treats all observations as independent and therefore ignores within-patient longitudinal dependence. Nevertheless, it provides a useful preliminary benchmark before introducing mixed-effects longitudinal modeling.

The fitted regression results are summarised below:

Term	Estimate	t-value	p-value
time	−2.01	−6.38	< 0.001
HTN	−1.59	−0.61	0.543
proteinuria	−6.21	−5.06	< 0.001
baseline eGFR (centered)	0.91	34.89	< 0.001
time × HTN	1.48	4.91	< 0.001
time × proteinuria	0.66	4.36	< 0.001
time × baseline eGFR	−0.027	−7.21	< 0.001

The model achieved an R^2 value of 0.758 with residual standard error 15.29. Interpreted directly, the negative coefficient for time suggests an average decline in eGFR over successive visits. Higher baseline proteinuria was associated with lower observed kidney function,

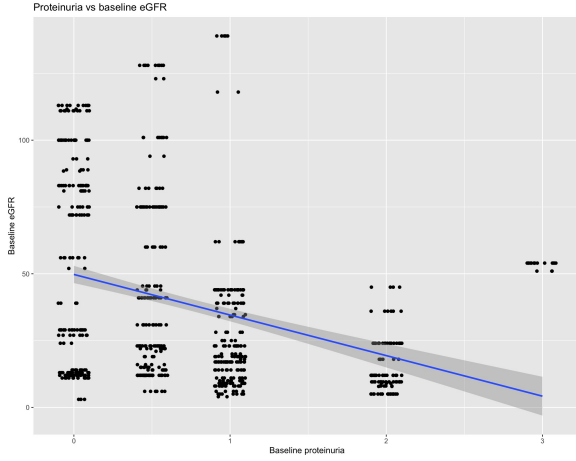


Figure 28: Baseline proteinuria versus baseline eGFR. The fitted line shows a clear negative association.

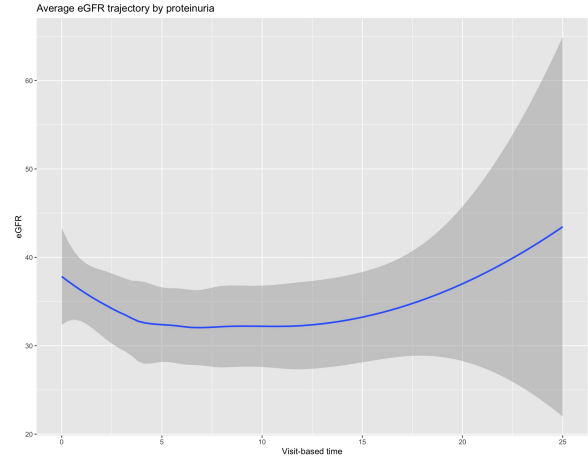


Figure 29: Smoothed average longitudinal eGFR trajectory over visit-based time.

while higher baseline eGFR strongly predicted higher subsequent eGFR measurements. The interaction terms additionally suggested that progression rates may vary according to baseline clinical status.

However, these results must be interpreted cautiously. The exploratory trajectory plots already demonstrate that longitudinal observations from the same patient are highly correlated and that progression slopes differ substantially across individuals. The naive regression framework ignores this dependence structure and therefore cannot adequately represent patient-specific longitudinal dynamics. Consequently, the next stage of the analysis replaced this benchmark model with a longitudinal mixed-effects growth model incorporating patient-specific random effects.

7.1 Longitudinal growth model for CKD progression

Building on the naive regression benchmark, the next step was to use a model that respects the repeated-measures structure of the data. The dataset contains multiple visits per patient, with highly unbalanced follow-up: some patients contribute only a few observations, while others are followed for more than 20 visits. Since the same patient appears repeatedly in the dataset, the observations are not independent, and a standard linear regression is not formally appropriate for inference on progression. The exploratory plots already suggest this clearly: individual eGFR trajectories are far from parallel, and the amount of follow-up varies substantially across patients.

Growth model specification

To model CKD progression properly, a linear mixed-effects growth model was fitted:

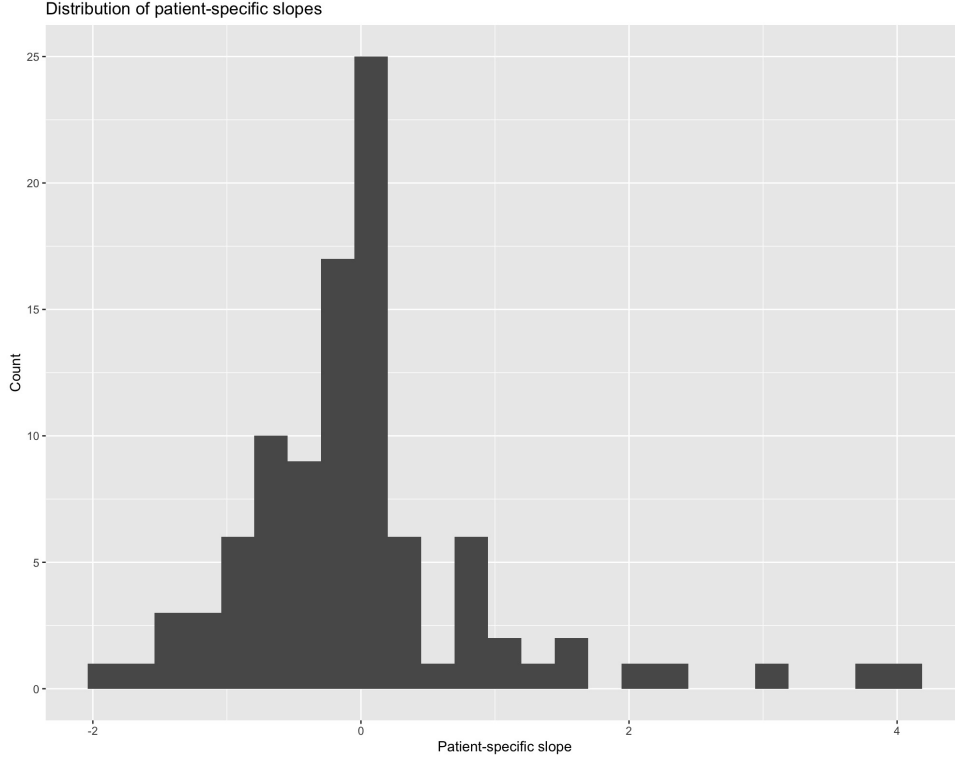


Figure 30: Smoothed average eGFR trajectory by proteinuria severity. The overall pattern suggests decline over time, but the widening confidence band at later visits reflects sparse data and increasing instability at long follow-up times.

$$\begin{aligned} \text{eGFR}_{ij} = & (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})t_{ij} + \beta_2\text{HTN}_i + \beta_3\text{Proteinuria}_i + \beta_4\text{Baseline eGFR}_{i,c} \\ & + \beta_5t_{ij}\text{HTN}_i + \beta_6t_{ij}\text{Proteinuria}_i + \beta_7t_{ij}\text{Baseline eGFR}_{i,c} + \varepsilon_{ij}, \end{aligned}$$

where b_{0i} is a patient-specific random intercept and b_{1i} is a patient-specific random slope. This formulation allows each patient to begin at a different baseline eGFR level and decline at a different rate. In this dataset, that flexibility is essential: the exploratory plots show that trajectories are clearly not parallel.

The fitted model was estimated by maximum likelihood and gave:

$$\text{AIC} = 6059.8, \quad \text{BIC} = 6115.3, \quad \log L = -3017.9.$$

The estimated random-effects variances were:

$$\text{Var}(b_{0i}) = 35.88, \quad \text{Var}(b_{1i}) = 2.42, \quad \text{Var}(\varepsilon_{ij}) = 141.39.$$

The random slope variance is clearly nonzero, which confirms that the data support

patient-specific decline rates rather than a single common slope. The negative intercept–slope correlation (-0.25) suggests that patients with higher baseline eGFR tend to decline somewhat faster, which is consistent with the significant baseline-eGFR-by-time interaction reported below.

Fixed effects and interpretation

The fixed-effects estimates were:

Term	Estimate	<i>p</i> -value
time	-1.290	0.082
HTN _{base}	-0.816	0.795
proteinuria _{base}	-3.791	0.009
baseline eGFR (centered)	0.919	< 0.001
time $\times$ HTN _{base}	1.157	0.111
time $\times$ proteinuria _{base}	-0.371	0.335
time $\times$ baseline eGFR (centered)	-0.0316	< 0.001

The main longitudinal finding is the strong negative interaction between time and baseline eGFR:

$$\beta_{\text{time} \times \text{baseline eGFR}} = -0.0316, \quad p < 0.001.$$

This means that patients with higher baseline kidney function tend to show a steeper subsequent decline in eGFR over visits. Quantitatively, a 10-unit increase in baseline eGFR corresponds to an additional decline of about 0.32 eGFR units per visit interval.

Proteinuria has a significant negative main effect (-3.791 , $p = 0.009$), showing that higher baseline proteinuria is associated with lower renal function at baseline. However, its interaction with time is not significant ($p = 0.335$), so the model does not provide strong evidence that proteinuria independently changes the rate of eGFR decline once patient-specific trajectory heterogeneity is accounted for. Hypertension is not significant either in the baseline effect or in the time interaction, which suggests that its apparent role in the simpler regression model was weakened after the longitudinal correlation structure was modeled properly.

The fitted time effect itself is negative (-1.290), but only marginally significant ($p = 0.082$), because much of the variation in progression is now absorbed by the random slope term. This is expected in a mixed model: once each patient is allowed to have their own slope, the population-average slope becomes a more conservative summary of progression.

Model checking

Residual diagnostics showed that the model captures the broad longitudinal structure of the data but is not perfect, which is typical for messy clinical data.

The residual-versus-fitted plot shows a noticeable spread increase at larger fitted values, together with some curved structure and a few extreme residuals. This suggests mild heteroscedasticity and the presence of outlying observations.

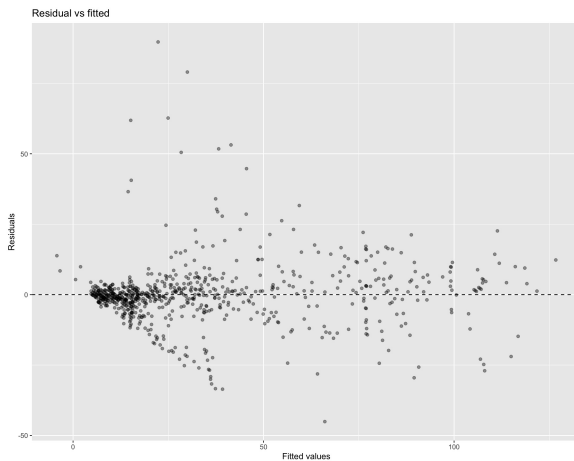


Figure 31: Residuals versus fitted values for the growth model. The residual spread increases at higher fitted values, indicating some heteroscedasticity and a few influential outliers.

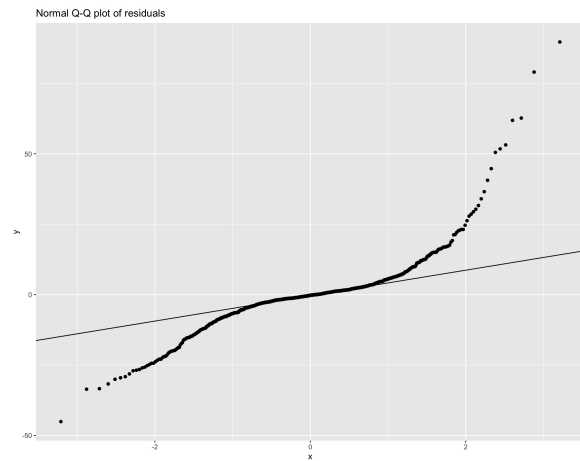


Figure 32: Normal Q–Q plot of residuals. The tail departures suggest non-normality and heavy-tailed residual behavior, which is common in noisy observational clinical data.

The normal Q–Q plot shows clear tail departures from normality, especially in the upper tail. The central part of the distribution is reasonably aligned, but the extremes deviate substantially, indicating heavy tails and a small number of influential points.

The observed-versus-predicted trajectory plot shows that the fitted blue lines broadly follow the patient-level data, which supports the overall adequacy of the random-slope structure. The model does not perfectly reproduce every abrupt fluctuation, but it captures the dominant pattern of longitudinal decline while allowing for substantial patient-specific variation.

Limitations of the analysis

Despite being more appropriate than ordinary linear regression, the mixed-effects growth model still has several limitations.

The follow-up structure is highly unbalanced: many patients contribute only a few visits, while a smaller subset contributes very long trajectories (Figure ??). Consequently, estimates at larger visit times are based on sparse information, which explains the widening

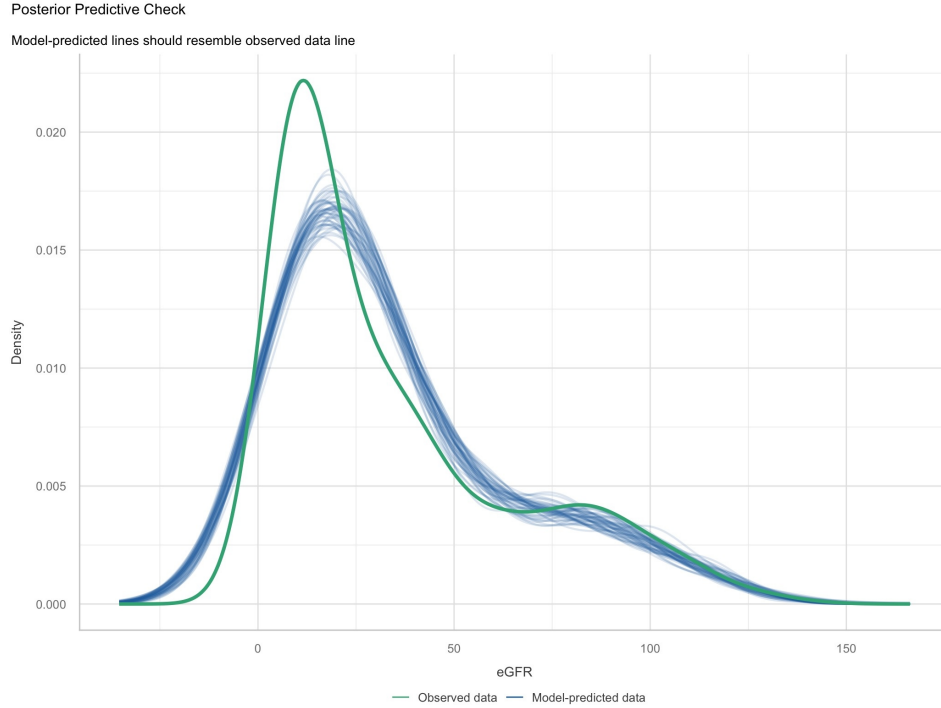


Figure 33: Observed and model-predicted eGFR trajectories. The fitted trajectories broadly track the observed longitudinal patterns while allowing patient-specific variation through random intercepts and slopes.

uncertainty bands in Figure 30.

Visit order was used as the time scale because calendar dates were incomplete and inconsistently recorded. Thus, one unit of “time” represents one additional recorded visit rather than a fixed duration. The estimated slopes should therefore be interpreted as change per visit, not change per year.

Residual diagnostics indicate departures from ideal model assumptions. The residual-versus-fitted plot (Figure 31) suggests heteroscedasticity, while the Q-Q plot (Figure 32) shows heavy-tailed residual behavior and outliers. Hence, the Gaussian assumptions underlying the model are only approximately satisfied.

The analysis also relies entirely on baseline covariates. Proteinuria and hypertension may vary substantially during follow-up, but reliable repeated measurements were not available for time-varying modelling. In addition, the model assumes approximately linear progression within each patient, whereas several observed trajectories suggest nonlinear decline patterns.

With more complete longitudinal data, stronger approaches would have been possible. Accurate calendar dates would allow modelling progression in real time, regular follow-up would permit non-linear mixed-effects models, and repeated covariate measurements would enable time-varying analyses. Information on clinical endpoints such as dialysis or

mortality would additionally allow joint longitudinal–survival modelling.

Nevertheless, the mixed-effects framework remains a substantial improvement over ordinary linear regression because it explicitly accounts for within-patient dependence and patient-specific progression trajectories.

Overall, the random-slope framework is a statistically more appropriate description of these data than the naive linear regression benchmark. The nonzero random slope variance confirms substantial heterogeneity in progression rates across patients.

Fixed effect	Estimate	<i>p</i>-value
time	−1.290	0.0819
HTN _{base}	−0.816	0.7949
proteinuria _{base}	−3.791	0.0091
baseline eGFR (centered)	0.9189	< 0.001
time × HTN _{base}	1.157	0.1106
time × proteinuria _{base}	−0.3707	0.3349
time × baseline eGFR (centered)	−0.0316	0.00035

Table 12: Fixed-effects estimates from the random-slope growth model.

The main longitudinal result was the significant negative interaction between time and baseline eGFR. Proteinuria was also significantly associated with lower baseline eGFR, but its interaction with time was not significant in this model.

As a secondary analysis, patient-specific slopes were estimated by fitting a simple linear model within each patient and then regressing those slopes on baseline covariates:

$$\widehat{\text{slope}}_i = \alpha + \gamma_1 \text{HTN}_i + \gamma_2 \text{Proteinuria}_i + \gamma_3 \text{Baseline eGFR}_{i,c} + \eta_i.$$

This model was jointly significant ($F = 8.548$, $p = 5.626 \times 10^{-5}$) and explained about 24.7% of the variation in patient-specific slopes ($R^2 = 0.247$, adjusted $R^2 = 0.219$). The results were consistent with the mixed model: proteinuria ($\hat{\beta} = -2.109$, $p < 0.001$) and baseline eGFR ($\hat{\beta} = -0.0436$, $p < 0.001$) were significant predictors of faster decline, whereas hypertension was not ($p = 0.586$).

7.2 Interaction effects in longitudinal progression

The same random-slope framework was extended by adding clinically motivated three-way interactions involving time. Since progression is represented by the slope of eGFR over visit-based time, the relevant terms are of the form $\text{time} \times A \times B$.

Interaction model	Three-way interaction term	p-value
HTN $\times$ Proteinuria	time:HTN_base:proteinuria_base	0.665
Proteinuria $\times$ Baseline eGFR	time:proteinuria_base:baseline_eGFR_c	0.00047
HTN $\times$ Baseline eGFR	time:HTN_base:baseline_eGFR_c	0.648

Table 13: Significance of three-way interaction terms in the longitudinal interaction models.

Interaction model	Interpretation
HTN $\times$ Proteinuria	No evidence that the longitudinal effect of proteinuria differs by hypertension status.
Proteinuria $\times$ Baseline eGFR	Strong evidence that the effect of proteinuria on progression depends on baseline kidney function. The interaction model also achieved the lowest AIC among the interaction models.
HTN $\times$ Baseline eGFR	No evidence that the effect of baseline kidney function on progression differs by hypertension status.

Table 14: Interpretation of the longitudinal interaction models fitted.

The hypertension-by-proteinuria model did not improve fit meaningfully over the additive growth model. The key three-way term was not significant, and the likelihood-ratio comparison also favoured the simpler model. This indicates that the longitudinal effect of proteinuria does not differ substantially between hypertensive and non-hypertensive patients.

The proteinuria-by-baseline-eGFR model was the only interaction model that significantly improved the analysis. Its three-way term was significant, and the model had the lowest AIC among the interaction models ($AIC = 6051.6$). This suggests that the prognostic effect of proteinuria depends on the stage of kidney function at baseline, with stronger progression-related information in patients with relatively preserved baseline eGFR.

The hypertension-by-baseline-eGFR model showed no meaningful improvement in fit and no significant three-way interaction. Thus, hypertension does not appear to modify the effect of baseline kidney function on longitudinal decline.

Overall, the interaction analysis indicates that most progression effects are adequately described by additive longitudinal associations. The only clearly supported effect modification is between proteinuria and baseline kidney function, which suggests that proteinuria is more informative as a progression marker in patients who start with better renal function.

8 Anaemia in ADPKD Across CKD Stages

Anaemia was evaluated in patients with Autosomal Dominant Polycystic Kidney Disease (ADPKD) using KDIGO-aligned sex-specific haemoglobin thresholds, defined as haemoglobin (Hb) < 13.5 g/dL in men and < 11.5 g/dL in women. The primary objective was to characterise the prevalence and severity of anaemia across progressive CKD stages (G1–G5), and to assess whether anaemia burden differed significantly by sex and renal function.

8.1 Overall prevalence of anaemia

Among female patients, 48 of 70 (68.6%) satisfied the anaemia criterion, whereas among male patients, 100 of 109 (91.7%) were anaemic. Overall, anaemia prevalence in the cohort was markedly high.

A Pearson chi-squared test with Yates' continuity correction demonstrated a statistically significant association between sex and anaemia prevalence:

$$\chi^2 = 14.406, \quad df = 1, \quad p = 0.0001473.$$

Thus, male patients with ADPKD exhibited a significantly greater prevalence of anaemia compared with female patients, despite the use of sex-specific diagnostic thresholds.

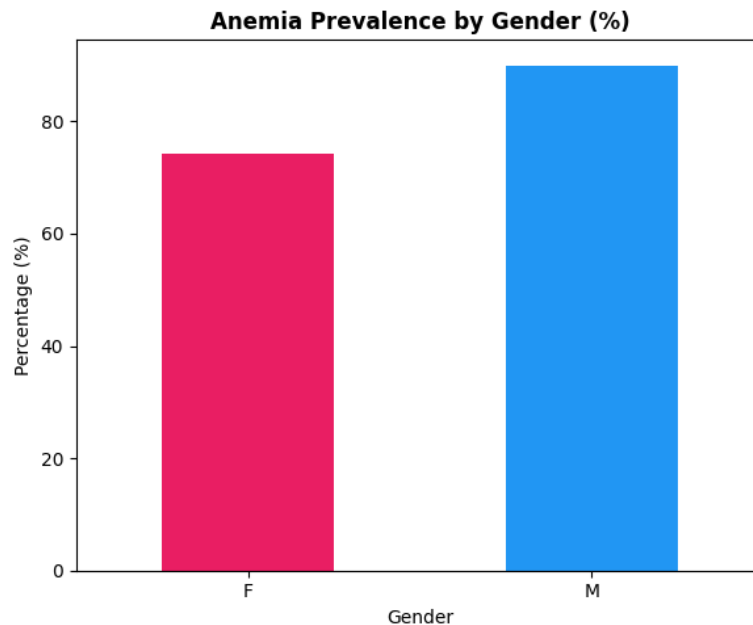


Figure 34: Anaemia prevalence by gender using KDIGO sex-specific haemoglobin thresholds.

The substantially elevated prevalence among men may reflect differences in disease severity,

renal dysfunction, erythropoietin deficiency, inflammatory burden, or baseline haemoglobin physiology. However, causal interpretation should be avoided because the present analysis does not adjust for potential confounding variables such as age, CKD stage distribution, iron status, or comorbidity burden.

8.2 Haemoglobin distribution and Variation of anaemia prevalence across CKD stages

The distribution of haemoglobin concentrations across CKD stages exemplifies the progressive worsening of anaemia severity with declining kidney function.

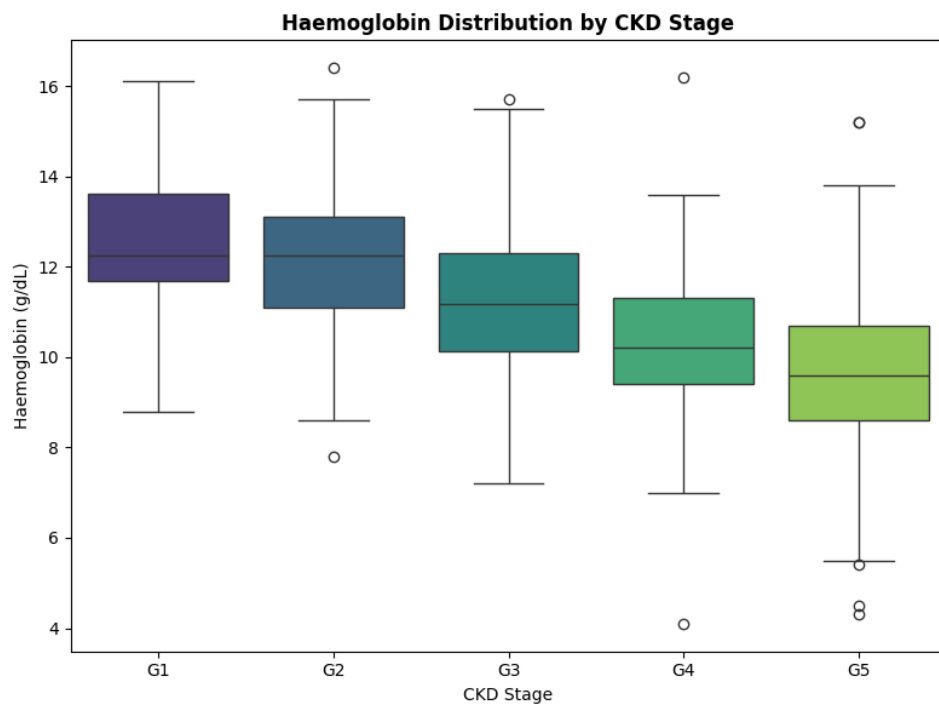


Figure 35: Distribution of haemoglobin levels across CKD stages.

Median haemoglobin levels declined progressively from G1 to G5, accompanied by increasing variability and lower-tail outliers in advanced stages. Patients in stages G4 and G5 exhibited substantially lower central tendencies and greater dispersion, indicating increasing heterogeneity in anaemia burden with advancing renal impairment.

Several severe low-haemoglobin observations were noted in stages G4 and G5, suggesting the emergence of clinically significant anaemia in advanced CKD. The downward shift in both median and interquartile range across stages is consistent with progressive failure of renal endocrine function.

The contingency table on anaemia prevalence further supported the progressive worsening of anaemia with declining kidney function. The stage-wise distribution is summarised in

Table 15.

Table 15: Anaemia prevalence across CKD stages

CKD Stage	Non-anaemic	Anaemic	Prevalence (%)
G1	10	6	37.5
G2	8	12	60.0
G3	6	24	80.0
G4	5	34	87.2
G5	2	72	97.3

A Pearson chi-squared test demonstrated a highly significant association between CKD stage and anaemia prevalence:

$$\chi^2 = 41.737, \quad df = 4, \quad p = 1.891 \times 10^{-8}.$$

These findings demonstrate a strong monotonic increase in anaemia prevalence with progressive renal dysfunction. The prevalence rose sharply after stage G2, with nearly universal anaemia observed among patients in stage G5 disease.

This pattern is biologically plausible and consistent with established CKD pathophysiology, whereby progressive nephron loss impairs renal erythropoietin production and contributes to chronic inflammation, iron dysregulation, reduced erythrocyte survival, and uraemic marrow suppression.

8.3 Severity distribution of anaemia

Anaemia severity was classified into mild, moderate, and severe categories according to KDIGO/WHO criteria. Overall, moderate anaemia constituted the majority of cases.

Table 16: Overall severity distribution of anaemia

Severity	Percentage (%)
Mild	31.1
Moderate	61.7
Severe	7.2

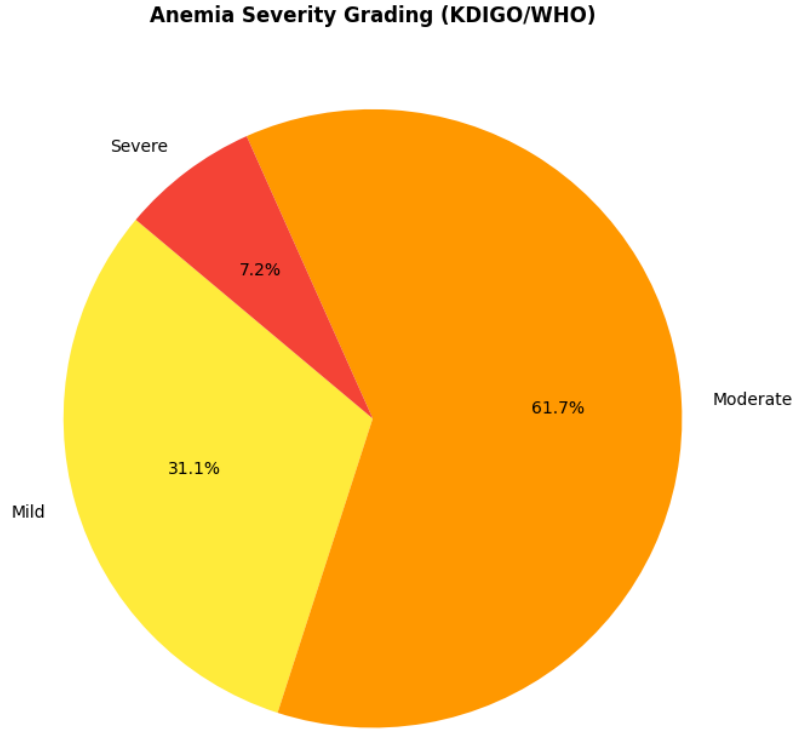


Figure 36: Overall severity grading of anaemia in ADPKD patients.

Moderate anaemia clearly predominated, whereas severe anaemia constituted a relatively smaller but clinically important subgroup.

8.4 Severity progression across CKD stages

The stage-wise severity distribution demonstrated progressive enrichment of moderate and severe anaemia with worsening CKD stage, as shown in Table 17.

Table 17: Anaemia severity across CKD stages

Severity	G1	G2	G3	G4	G5
Mild	3	9	12	13	13
Moderate	3	3	10	20	48
Severe	0	0	2	1	11

A Fisher's exact test with Monte Carlo simulation demonstrated a statistically significant

association between CKD stage and anaemia severity distribution:

$$p = 0.0019.$$

This result indicates that the composition of anaemia severity changes systematically across CKD stages. In early CKD (G1–G2), mild anaemia predominated, whereas moderate anaemia became increasingly dominant in G4–G5 disease. Severe anaemia was largely concentrated in advanced CKD, particularly stage G5.

Importantly, the proportion of moderate-to-severe anaemia increased substantially with worsening renal function, suggesting not only a greater prevalence of anaemia in advanced CKD but also a clinically meaningful intensification in severity.

8.5 Discussion

Taken together, the analyses demonstrate three principal findings:

1. Anaemia is highly prevalent among patients with ADPKD, particularly in advanced CKD stages.
2. Both prevalence and severity of anaemia increase significantly with worsening renal function.
3. Male patients exhibited significantly higher anaemia prevalence despite the use of sex-specific KDIGO diagnostic thresholds.

The strong stage-wise progression observed here is clinically important because anaemia in CKD is associated with reduced quality of life, cardiovascular morbidity, exercise intolerance, faster disease progression, and increased mortality risk.

The predominance of moderate anaemia in advanced stages suggests that many patients may require closer monitoring for erythropoietin deficiency, iron dysregulation, and consideration of anaemia-directed therapies.

8.6 Statistical considerations

The analyses were appropriately based on categorical association testing. Pearson chi-squared tests were used for prevalence comparisons where expected counts were adequate, whereas Fisher’s exact test with Monte Carlo simulation was used for anaemia severity because sparse cell counts were present in severe categories.

However, several limitations should be acknowledged. The analyses are cross-sectional and therefore do not establish temporal causality. Potential confounders such as age, iron status, inflammation, dialysis exposure, medication use, and nutritional status were not

adjusted for. Furthermore, CKD stage is ordinal, and future analyses could employ ordinal logistic regression or Cochran–Armitage trend testing to quantify monotonic progression more formally.

Despite these limitations, the observed statistical associations are strong, clinically coherent, and biologically plausible.

9 Conclusion

To sum up, our study investigated longitudinal CKD progression in a retrospective Indian ADPKD cohort consisting of 193 patients and 982 follow-up observations. Extensive preprocessing was required before statistical analysis. Clinical variables recorded in heterogeneous free-text formats were standardized into analytically usable forms. Multiple imputation using MICE with Predictive Mean Matching (PMM) was employed to address structured missingness while preserving biologically plausible distributions.

- Exploratory analysis demonstrated that the cohort presented at an unusually advanced stage of disease compared with international ADPKD registries. Median baseline eGFR was approximately 19 mL/min/1.73m², more than 40% of patients were already in CKD stage G5, and hypertension prevalence exceeded 85%. Progressive worsening of biochemical abnormalities, particularly anaemia and renal dysfunction, was observed with advancing CKD stage.
- Survival analysis was performed using a composite renal endpoint defined by either a $\geq 40\%$ decline in eGFR or progression to ESKD. Kaplan–Meier estimation demonstrated steadily declining progression-free survival over follow-up, with nearly half the cohort experiencing clinically meaningful progression by approximately 12 visits. Baseline kidney function showed the clearest and most statistically significant separation in progression-free survival curves, while proteinuria demonstrated a clinically suggestive dose-response relationship with worsening outcomes.
- Cox proportional hazards regression was used to evaluate baseline determinants of renal progression. Lower baseline eGFR emerged as the strongest predictor of increased progression risk after accounting for demographic and clinical covariates. Other variables including hypertension, haemoglobin, uric acid, and cardiovascular disease demonstrated clinically plausible trends but generally lacked strong statistical significance, likely due to limited sample size, sparse measurements, and cohort imbalance.
- Longitudinal renal decline was further characterized through patient-specific estimation of annualized eGFR slopes using ordinary least squares regression. The

median annualized decline rate was approximately $-1.73 \text{ mL/min/1.73m}^2/\text{year}$, although substantial heterogeneity in progression trajectories was observed across individuals. Older age, lower baseline eGFR, and history of urinary tract infections were associated with more rapid renal decline. A statistically significant interaction between age and baseline kidney function suggested that the effect of renal reserve on progression may differ across age groups.

- Dedicated analysis of anaemia demonstrated extremely high prevalence and progressively increasing severity with worsening CKD stage. Moderate anaemia predominated overall, while severe anaemia was concentrated almost entirely among advanced CKD patients. Male patients exhibited significantly higher anaemia prevalence despite the use of sex-specific diagnostic thresholds, suggesting substantial systemic disease burden within the cohort.

Overall, the study demonstrated that Indian ADPKD patients frequently present with markedly advanced disease and substantial comorbidity burden at first nephrology contact. Baseline kidney function consistently emerged as the strongest determinant of subsequent progression, whereas proteinuria and hypertension showed clinically important but statistically unstable associations owing to extensive missingness and limited statistical power. The findings emphasize the need for earlier diagnosis, structured longitudinal registries, imaging-based risk stratification, and prospective multicentre studies incorporating genetic and radiological biomarkers.

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